

EX NO: 1	LIBRARY MANAGEMENT SYSTEM
DATE:	

AIM:

To draw the diagrams [use case, activity, sequence, collaboration, class, statechart, component, deployment, package] for the Library management system.

SOFTWARE REQUIREMENTS SPECIFICATION

SL.NO	SOFTWARE REQUIREMENTS SPECIFICATION
1.0	Hardware Requirements
1.1	Software Requirements
1.2	Problem Analysis and Project Plan
1.3	Project Description
1.4	Reference

1.0HARDWARE REQUIREMENTS:

- Intel Pentium Processor I3/I5
- Minimum 4 GB RAM
- Hard Disk: 500 GB

1.1 SOFTWARE REQUIREMENTS:

Operating System: Windows 7/10/11

UML Tool: Rational Rose / Argo UML

1.2PROBLEM ANALYSIS AND PROJECT PLANNING

The Library Management System is designed to simplify the process of managing books in a library. Users can search for books, borrow them, and return them. The librarian manages all operations such as issuing books, returning books, and maintaining records. The system ensures that book availability and user details are properly stored and updated.

1.3 PROJECT DESCRIPTION:

This software is developed to maintain the complete details of books and users in a library.

The system allows:

- Searching for books
- Issuing books to users
- Returning books
- Maintaining records

The librarian verifies the user details and updates the database accordingly.
This system reduces manual work and improves efficiency.

1.4 REFERENCES:

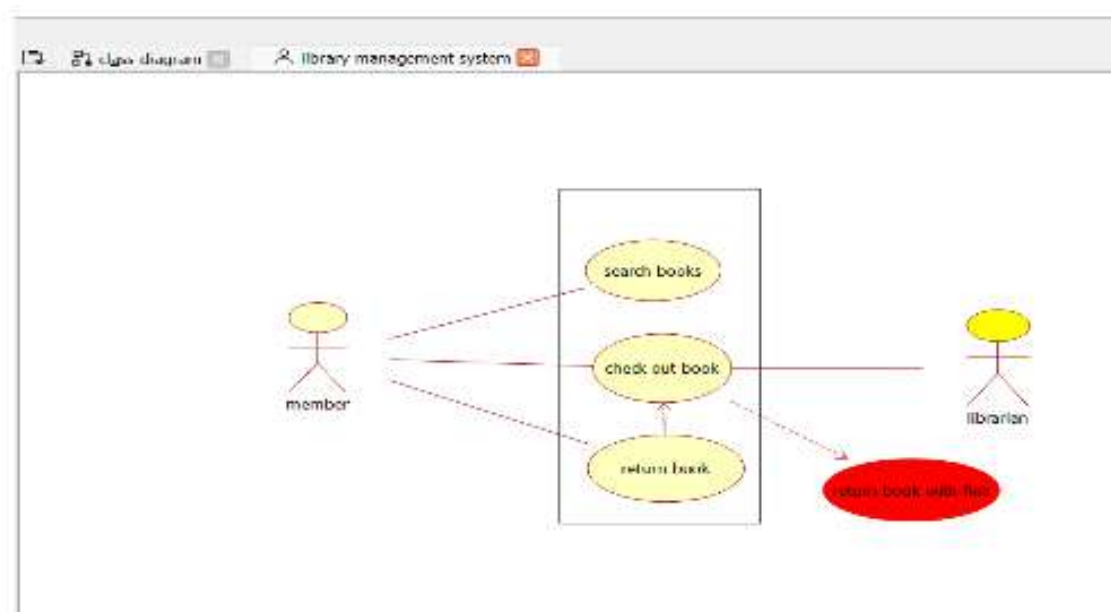
IEEE Software Requirement Specification format

USE CASE DIAGRAM:

This diagram will contain the actors, use cases which are given below

Actors: Student/User, Librarian

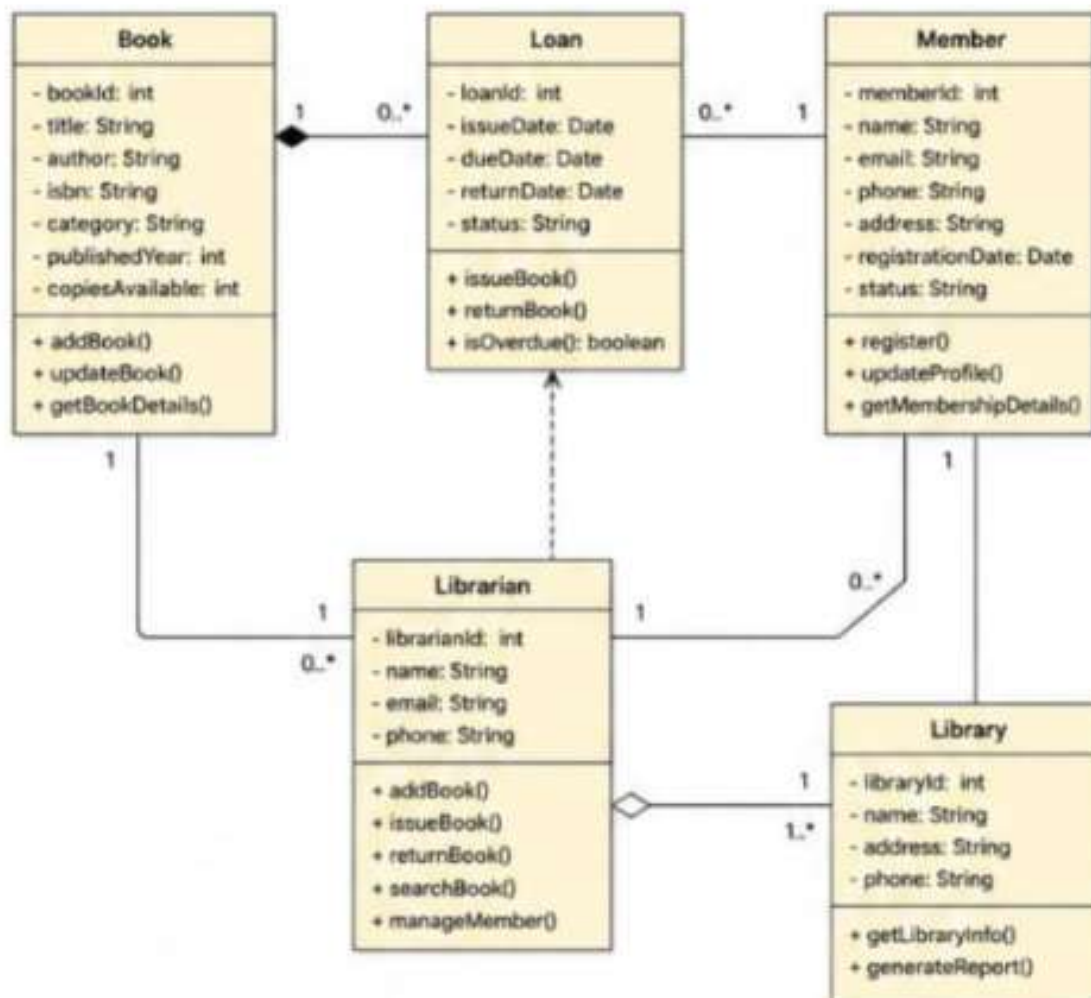
Use case: Search book, view book details, request book, issue of book, return of book, renewal of book, check availability, maintain records



CLASS DIAGRAM:

This diagram consists of the following classes, attributes and their operations

CLASSES	ATTRIBUTES	OPERATIONS
Library system	Book and user details	Check availability(), Maintain records()
User (Student)	Name, ID, Book issued	Search book(), Request book(), Return book()
librarian	Book records, user records	Issue book(), Return book(), Update records()



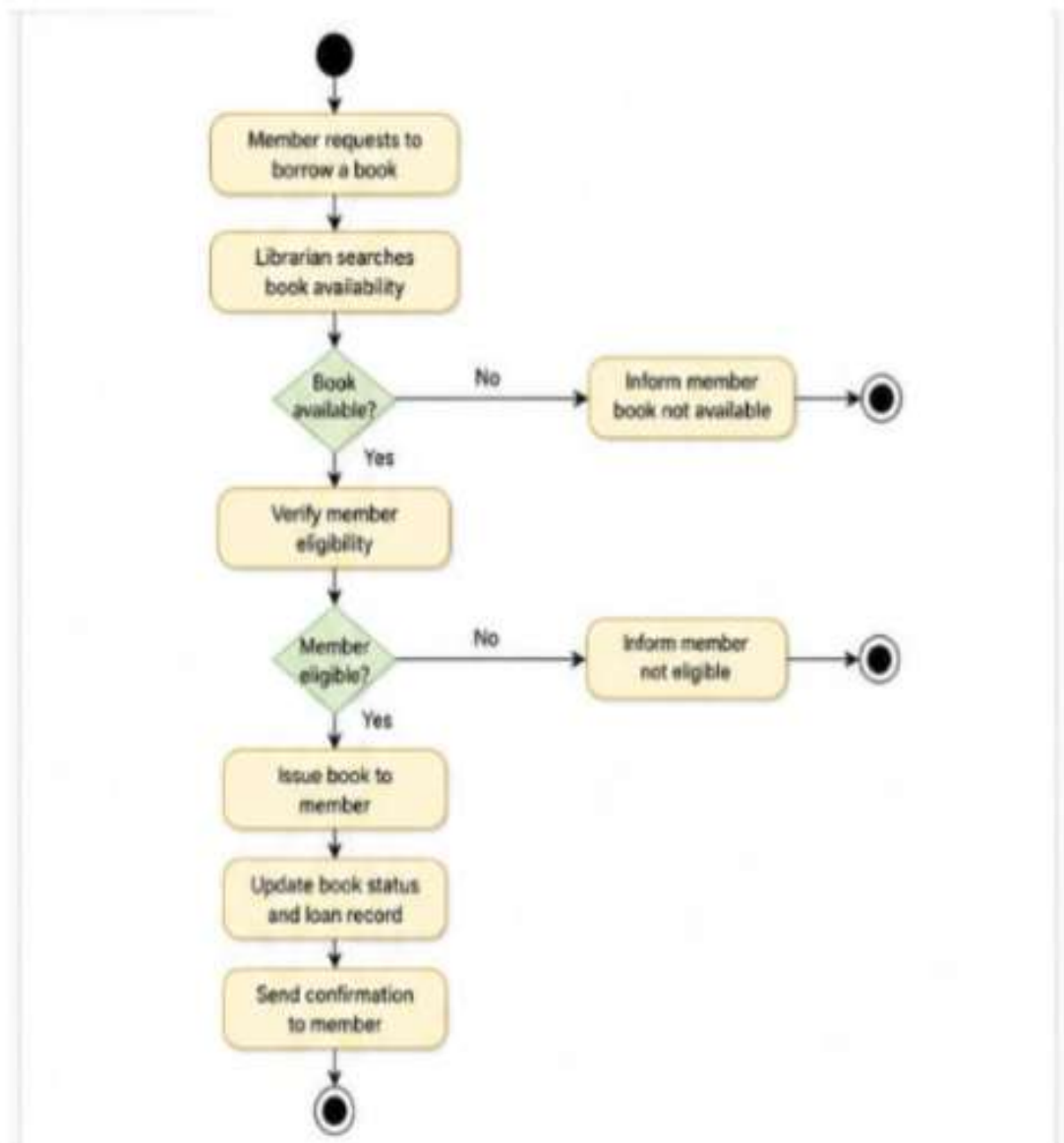
ACTIVITY DIAGRAM:

This diagram will have the activities as Start point, End point, Decision boxes as given

Below:

Activities: Enter applicant details , submission of proof, verification of books, issue books

Decision box: check the books availability

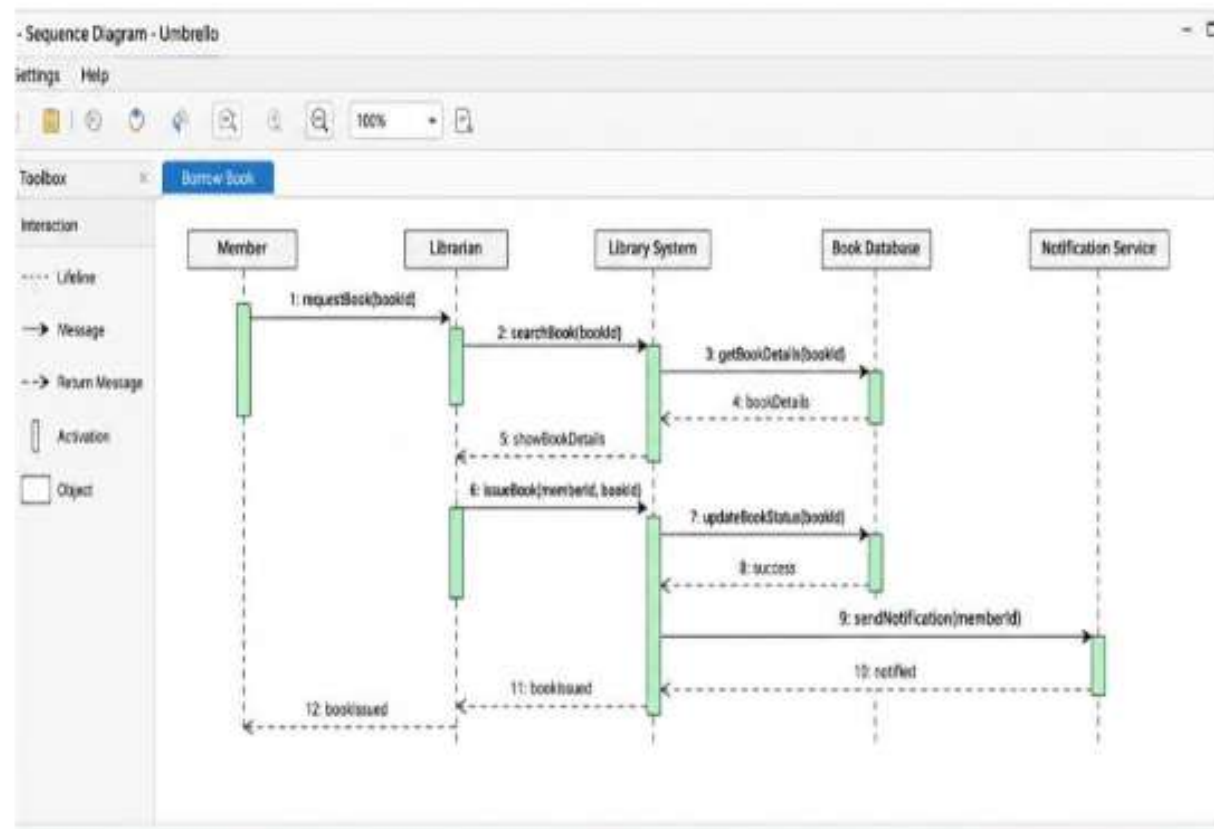


SEQUENCE DIAGRAM:

This diagram consists of the objects, messages, and return messages.

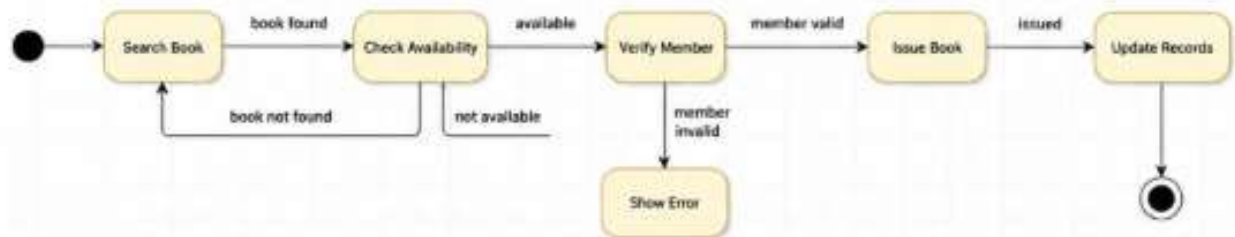
Objects:

Member/User, Librarian, Library Management System



STATE CHART DIAGRAM:

The purpose of a state chart diagram is to understand the process involved in performing a method in the **Library Management System**. It is also called a state diagram. A state is represented as a rounded box, which may contain one or more compartments. An initial state is represented as a small dot, and a final state is represented as a circle surrounding a small dot.

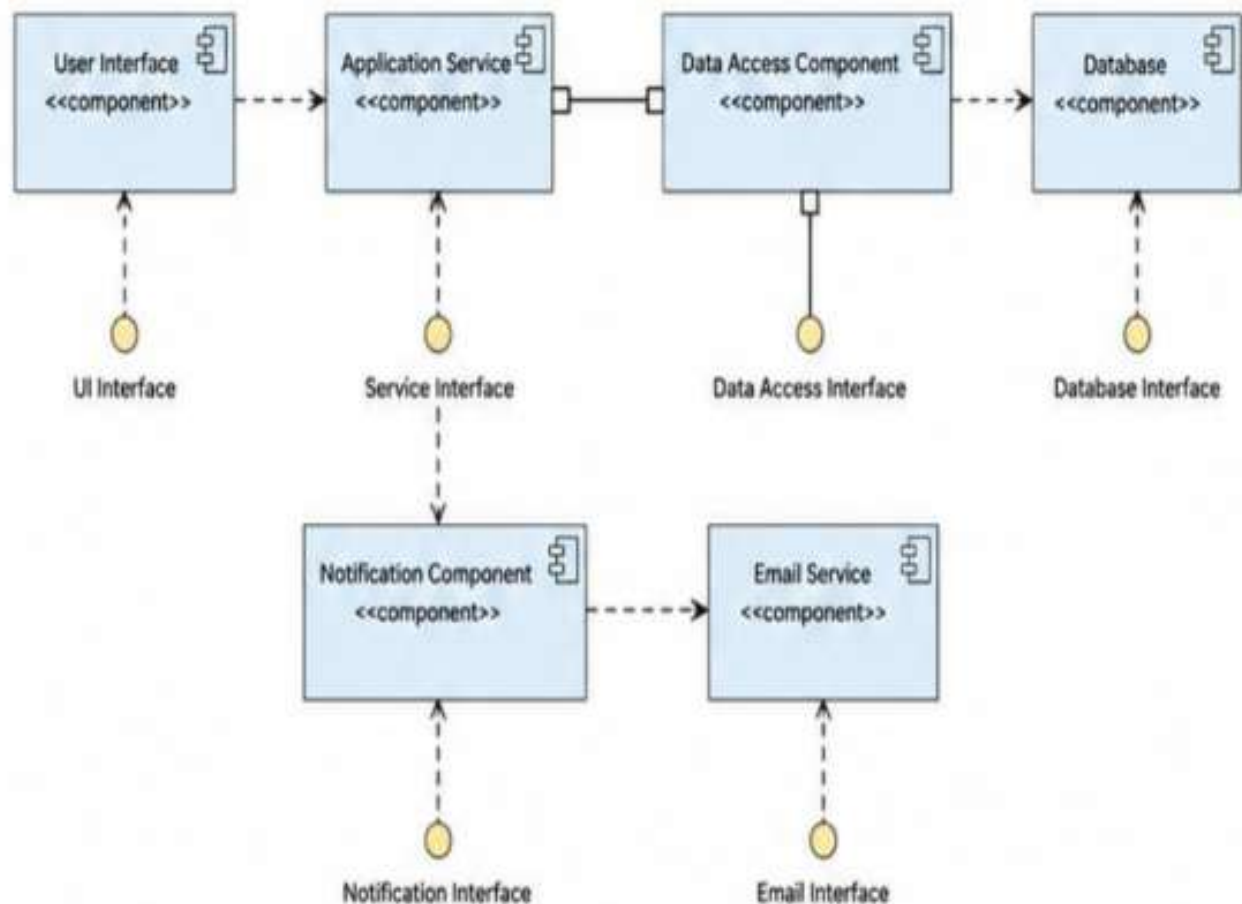


COMPONENT DIAGRAM:

The main purpose of the component diagram is to show the structural relationships between the components of the **Library Management System**. It is represented by boxed figures, and dependencies are represented by communication associations.

Components:

User Interface, Book Management Module, Member Management Module, Transaction Module, Database

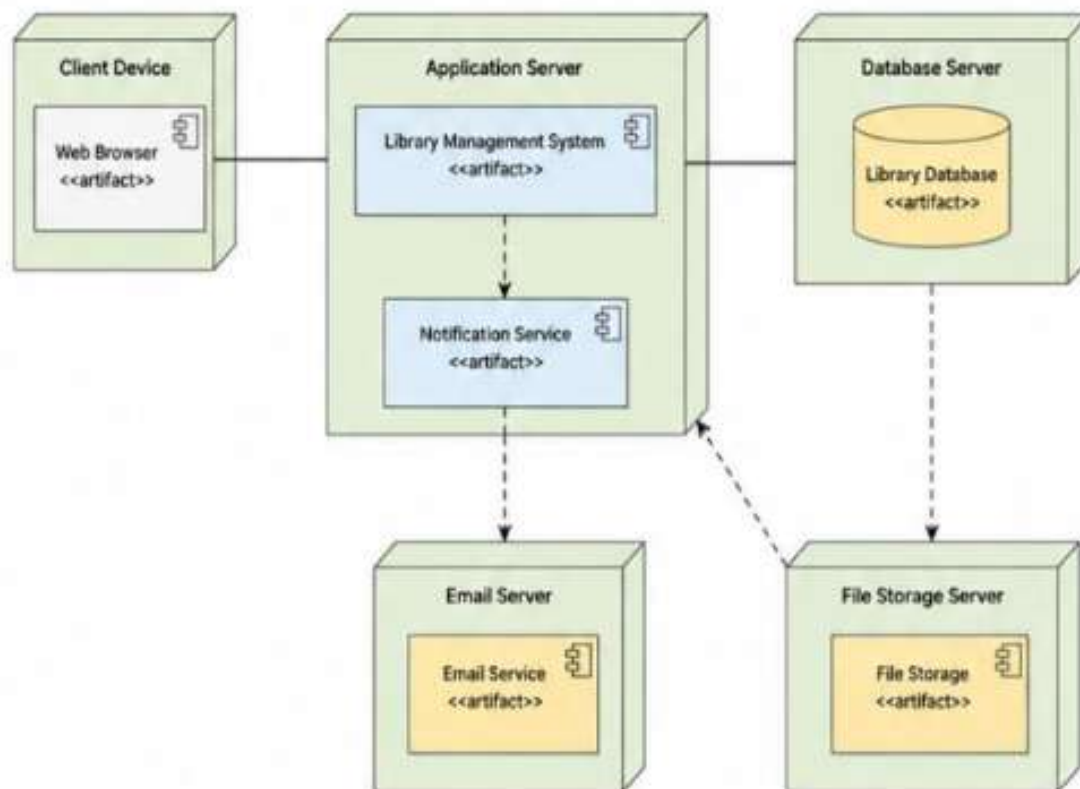


DEPLOYMENT DIAGRAM:

A deployment diagram in the Unified Modeling Language is used to model the physical deployment of artifacts in the **Library Management System**. It shows the allocation of artifacts to nodes according to the deployments defined between them. It is represented by a 3-dimensional box, and dependencies are represented by communication associations.

Nodes:

Client System, Application Server, Database Server



RESULT:

Thus the diagrams [use case, activity, sequence, collaboration, class, collaboration, component, deployment, package] for the Library Management System has been designed, executed and output is verified

EX NO: 2	RESTAURANT MANAGEMENT SYSTEM
DATE:	

AIM:

To draw the diagrams [use case, activity, sequence, collaboration, class, statechart, component, deployment, package] for the Restaurant Management System.

SOFTWARE REQUIREMENTS SPECIFICATION

SL.NO	SOFTWARE REQUIREMENTS SPECIFICATION
1.0	Hardware Requirements
1.1	Software Requirements
1.2	Problem Analysis and Project Plan
1.3	Project Description
1.4	Reference

1.5HARDWARE REQUIREMENTS:

- Intel Pentium Processor I3/I5
- Minimum 4 GB RAM
- Hard Disk: 500 GB

1.6 SOFTWARE REQUIREMENTS:

Operating System: Windows 7/10/11

UML Tool: Rational Rose / Argo UML

1.7PROBLEM ANALYSIS AND PROJECT PLANNING

The Restaurant Management System is designed to simplify restaurant operations. Customers can view menu items, place orders, and make payments.

The manager/staff handles order processing, billing, table management, and inventory updates.
The system ensures smooth restaurant operations and reduces manual work.

1.8 PROJECT DESCRIPTION:

This software is developed to maintain complete details of restaurant operations.

The system allows:

- Viewing menu items
- Placing food orders
- Managing tables
- Generating bills
- Processing payments
- Maintaining inventory records

The staff verifies orders and updates the database accordingly.

This system improves efficiency and customer service.

1.9 REFERENCES:

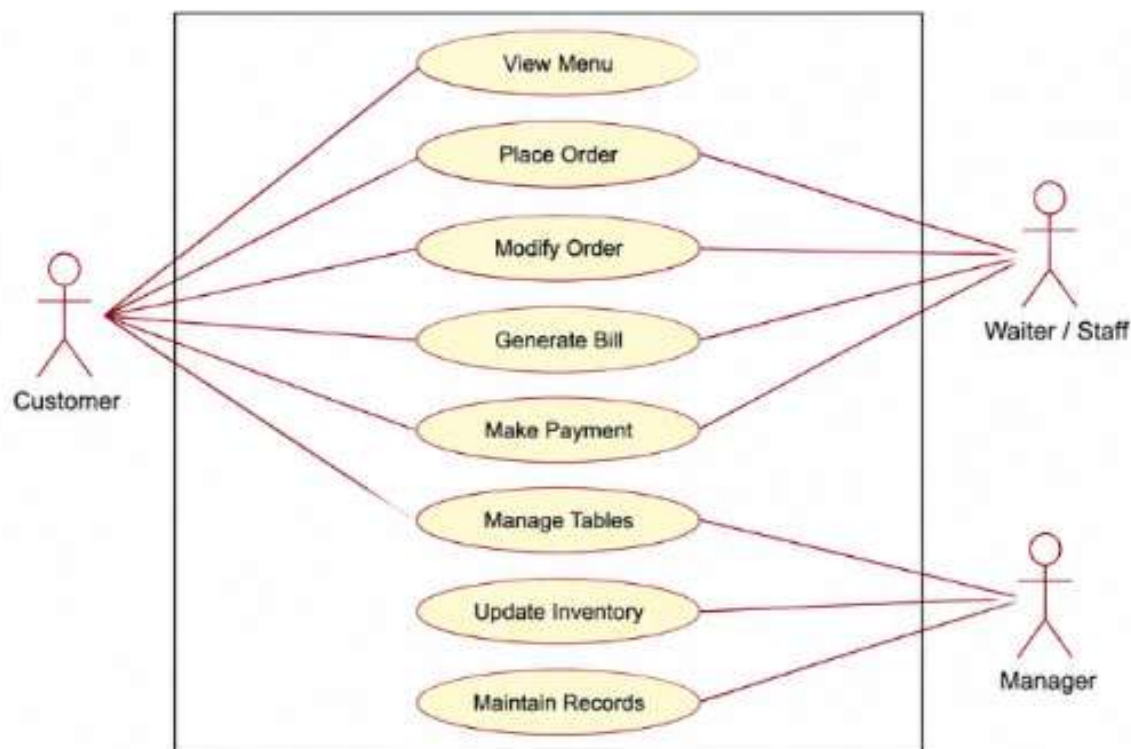
IEEE Software Requirement Specification format

USE CASE DIAGRAM:

Actors: Customer, Waiter/Staff, Manager

Use Cases (Horizontal Words):

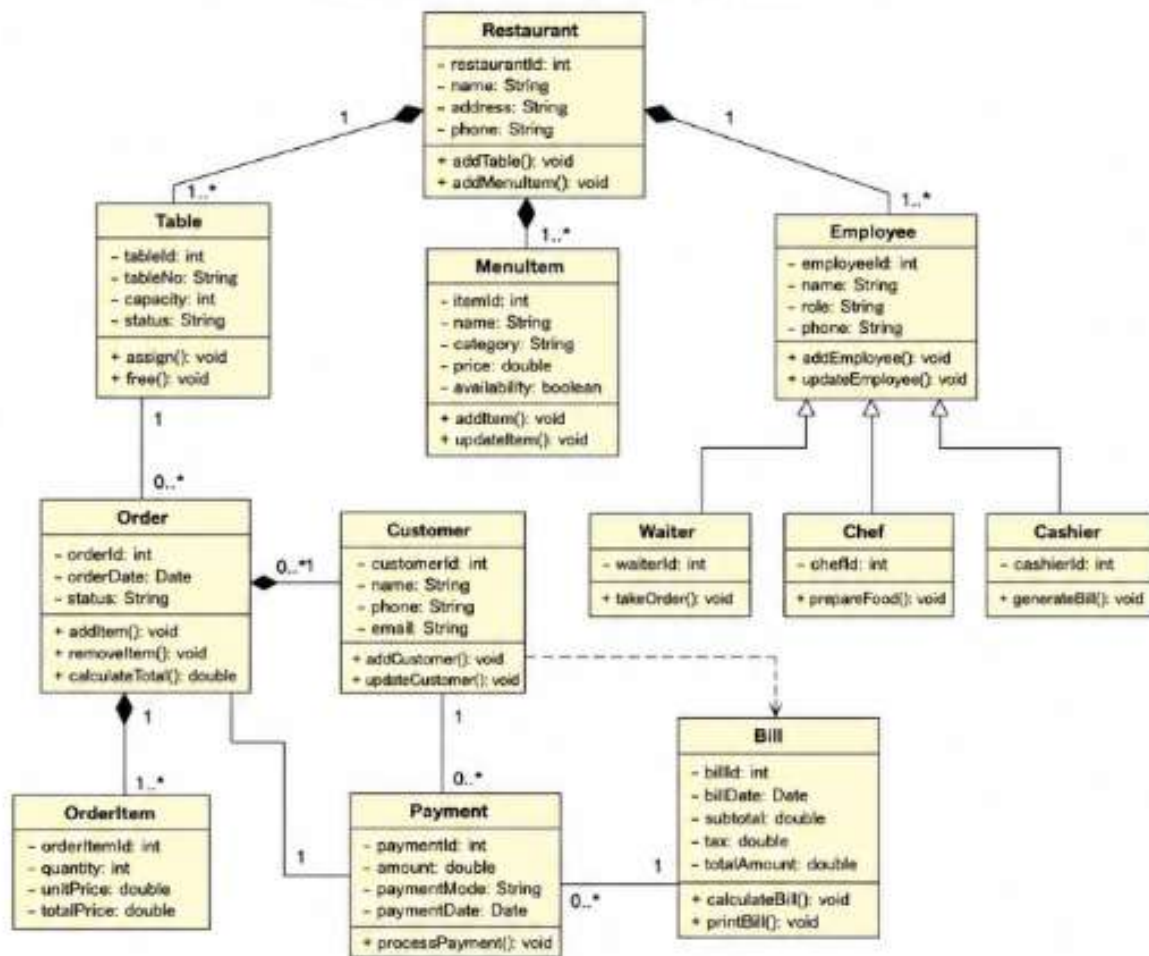
View Menu | Place Order | Modify Order | Generate Bill | Make Payment | Manage Tables | Update Inventory | Maintain Records



CLASS DIAGRAM:

This diagram consists of the following classes, attributes and their operations

CLASSES	ATTRIBUTES	OPERATIONS
Restaurant System	Menu details, Order details, Customer details	Manage orders(), Maintain records()
Customer	Name, Customer ID, Contact details, Order details	View menu(), Place order(), Make payment()
Chef	Food items, Order list, Preparation status	Prepare food(), Update order status()
Cashier	Bill details, Payment records	Generate bill(), Accept payment()
Manager	Staff details, Inventory details, Sales records	Update menu(), Manage staff(), View reports()



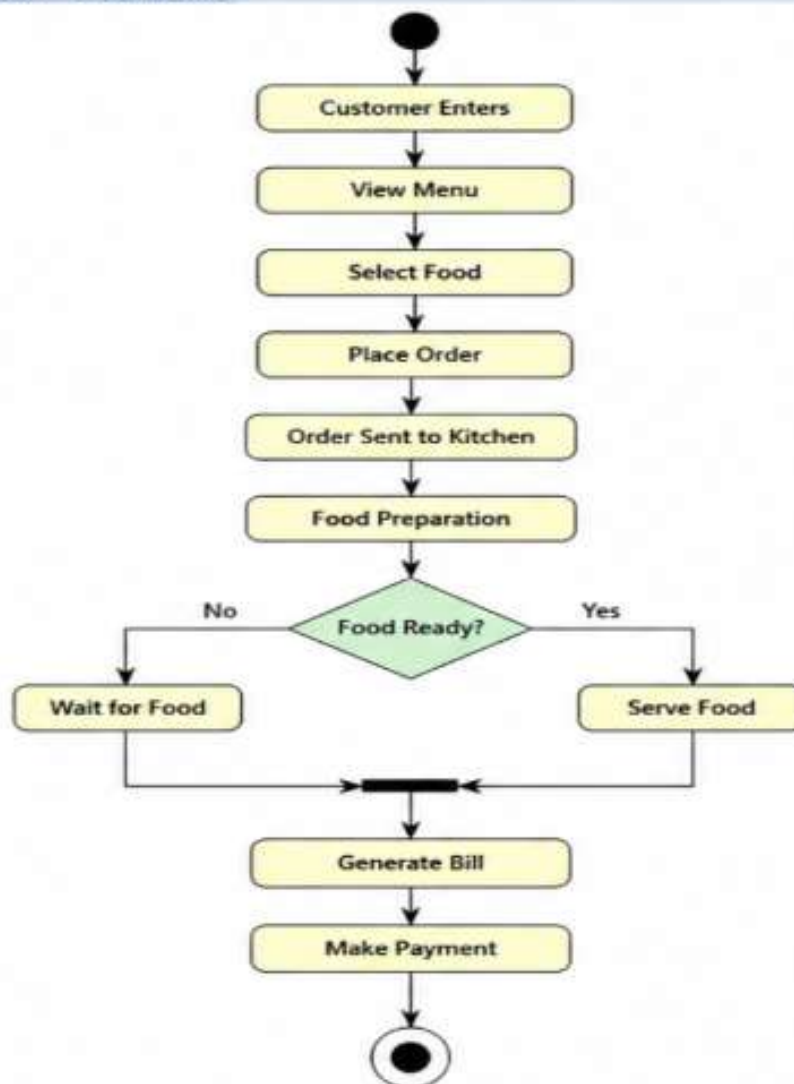
ACTIVITY DIAGRAM:

This diagram will have the activities such as Start point, End point, and Decision boxes as given below:

Activities: View menu, Select food items, Place order, Prepare food, Serve food, Generate bill, Make payment

Decision Box: Check food availability / Payment method verification

Activity Diagram - Order Activity



STATE CHART DIAGRAM (Restaurant Management System):

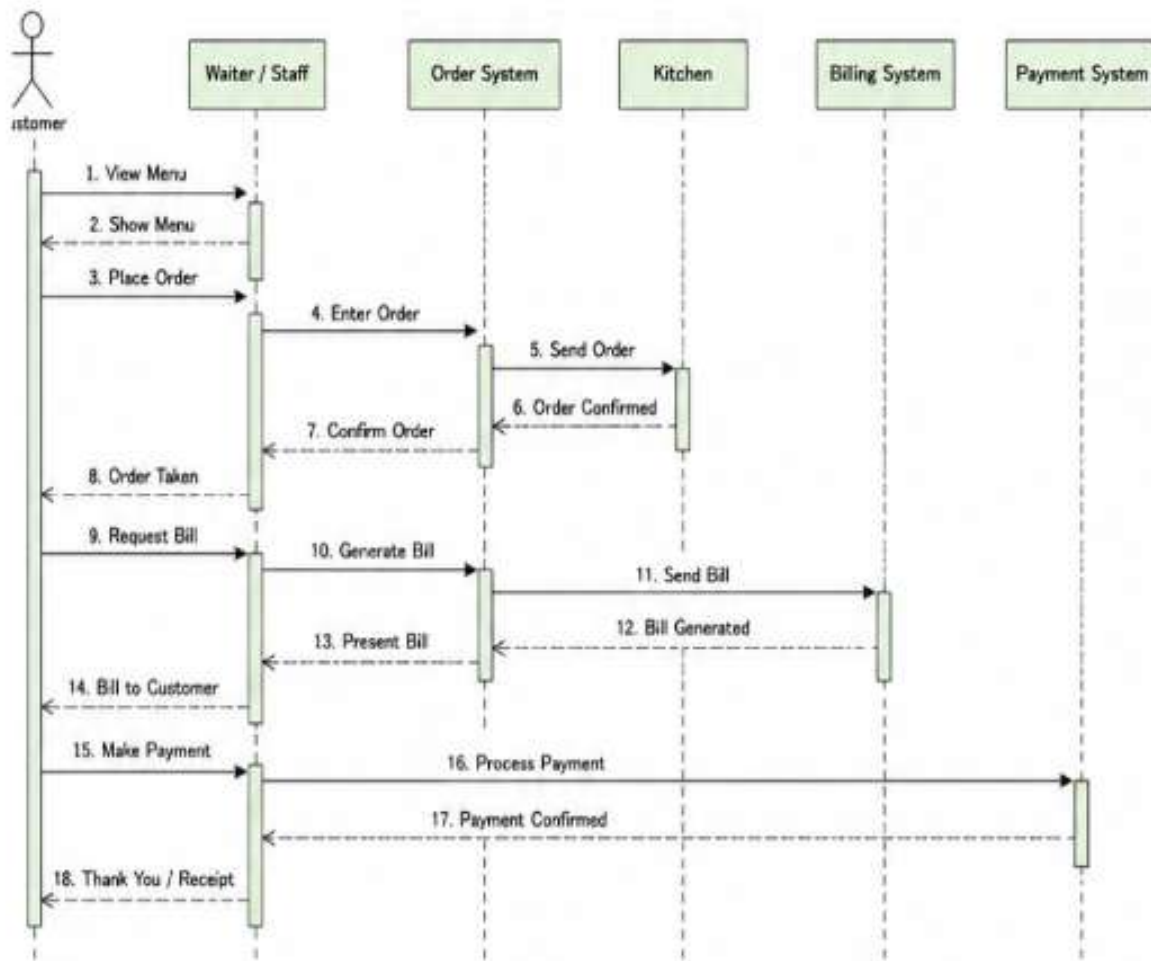
The purpose of a **state chart diagram** is to understand the different states involved in processing an order in the **Restaurant Management System**. It is also called a **state diagram**. A **state** is represented as a rounded box, which may contain one or more compartments. An **initial state** is represented as a small dot, and a **final state** is represented as a circle surrounding a small dot.

State Diagram - Order State



SEQUENCE DIAGRAM (Restaurant Management System):

This diagram consists of the **objects**, **messages**, and **return messages**.



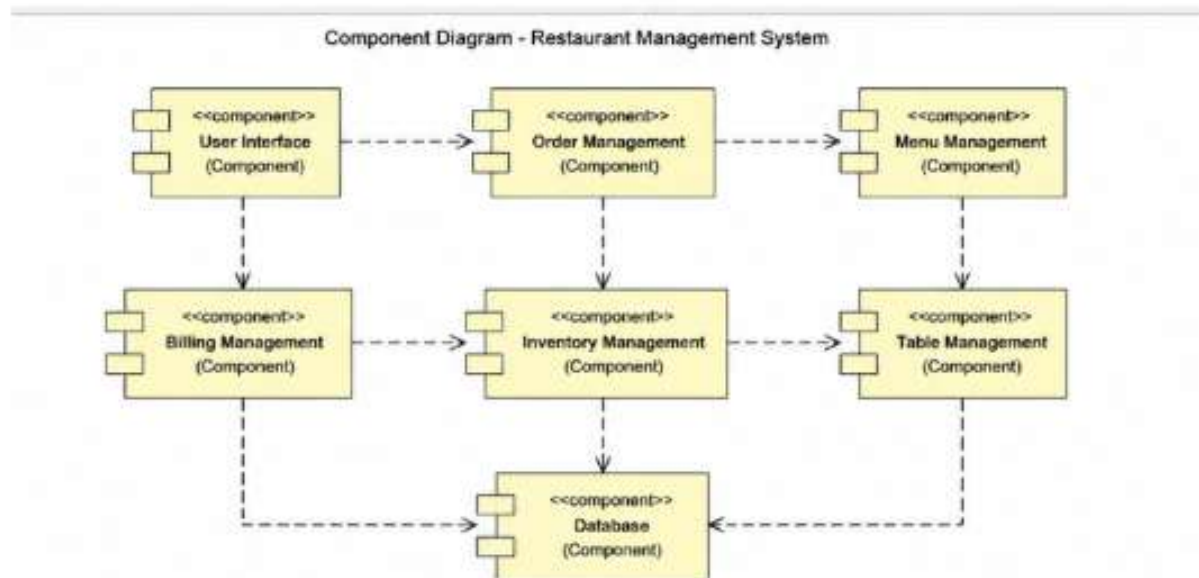
COMPONENT DIAGRAM:

The main purpose of the component diagram is to show the structural relationships between the components of the **Restaurant Management System**. It is represented by boxed figures, and dependencies are represented by communication associations.

Components:

- User Interface

- Menu Management Module
- Order Management Module
- Payment Module
- Kitchen Module
- Database



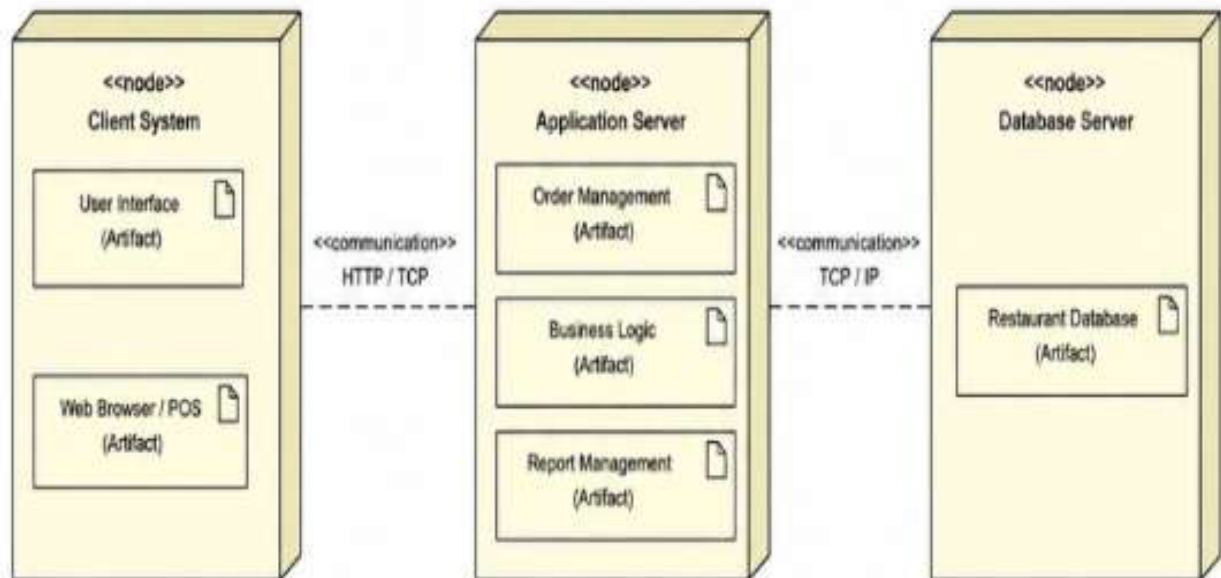
DEPLOYMENT DIAGRAM:

A deployment diagram in the Unified Modeling Language is used to model the physical deployment of artifacts in the **Restaurant Management System**. It shows the allocation of artifacts to nodes according to the deployments defined between them. It is represented by a 3-dimensional box, and dependencies are represented by communication associations.

Nodes:

- Customer/Admin System
- Application Server
- Database Server

Deployment Diagram - Restaurant Management System



RESULT:

Thus the diagrams [use case, activity, sequence, collaboration, class, collaboration, component, deployment, package] for the Library Management System has been designed, executed and output is verified

EX NO: 3	ONLINE VOTING SYSTEM
DATE:	

AIM:

To draw the diagrams [use case, activity, sequence, collaboration, class, state chart, component, deployment, package] for the Online Voting System.

SOFTWARE REQUIREMENTS SPECIFICATION

SL.NO	SOFTWARE REQUIREMENTS SPECIFICATION
1.0	Hardware Requirements
1.1	Software Requirements
1.2	Problem Analysis and Project Plan

SL.NO	SOFTWARE REQUIREMENTS SPECIFICATION
1.3	Project Description
1.4	Reference

1.0 HARDWARE REQUIREMENTS:

Intel Pentium Processor i3/i5

1.1 SOFTWARE REQUIREMENTS:

Rational Rose / Argo UML

1.2 PROBLEM ANALYSIS AND PROJECT PLAN

To simplify the voting process, software has been created using Rational Rose tool. Initially, the voter logs into the online voting system and submits voter details. These details are stored in the database and verified by the election administrator. After successful verification, the voter can cast the vote securely and the result will be stored in the central database.

1.3 PROJECT DESCRIPTION:

This software is designed for secure online voting and voter verification through a central server. The voter details are verified by the election authority before allowing voting access. The system ensures secure vote casting, maintains voter privacy, and generates election results accurately.

1.2 PROBLEM ANALYSIS AND PROJECT PLAN

To simplify the voting process, software has been created using Rational Rose tool. Initially, the voter logs into the online voting system and submits voter details. These details are stored in the database and verified by the election administrator. After successful verification, the voter can cast the vote securely and the result will be stored in the central database.

1.3 PROJECT DESCRIPTION:

This software is designed for secure online voting and voter verification through a central server. The voter details are verified by the election authority before allowing voting access. The system ensures secure vote casting, maintains voter privacy, and generates election results accurately.

1.4 REFERENCES:

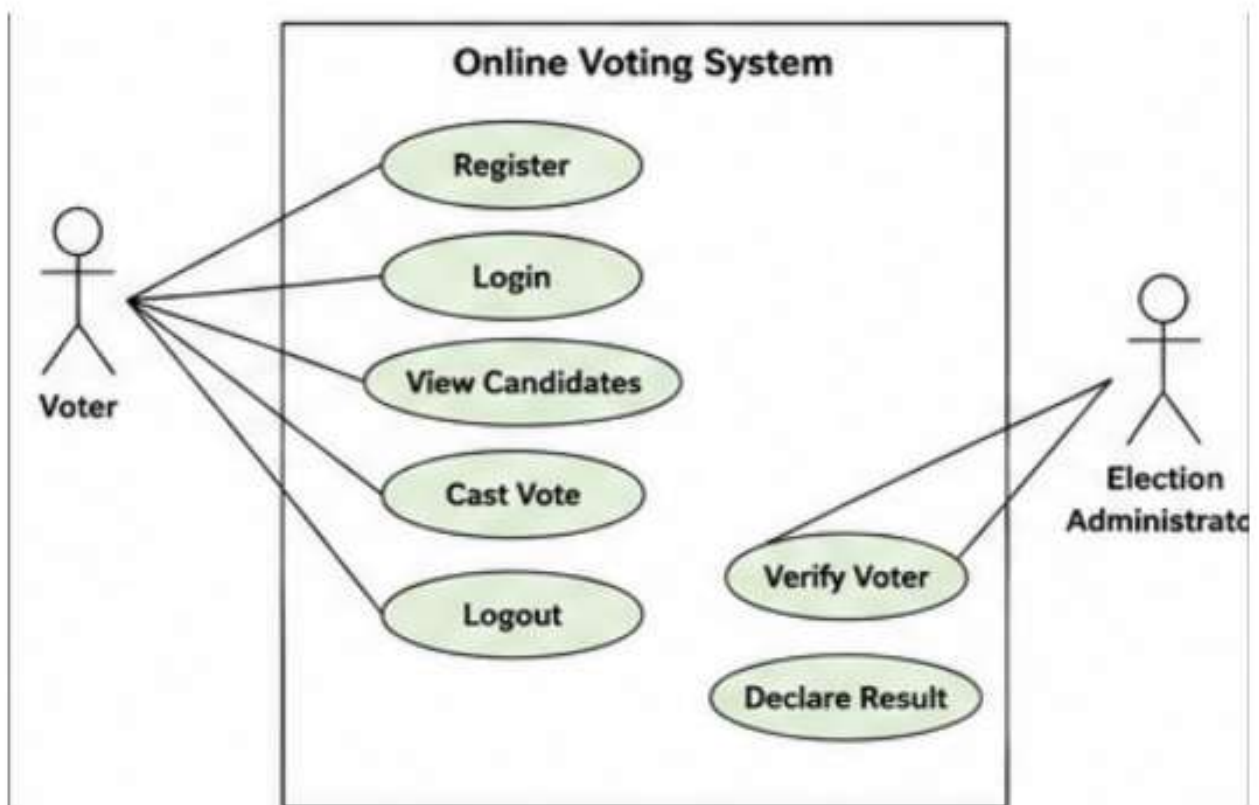
IEEE Software Requirement Specification format.

USE CASE DIAGRAM:

This diagram will contain the actors and use cases which are given below.

Actors: Voter, Election Administrator.

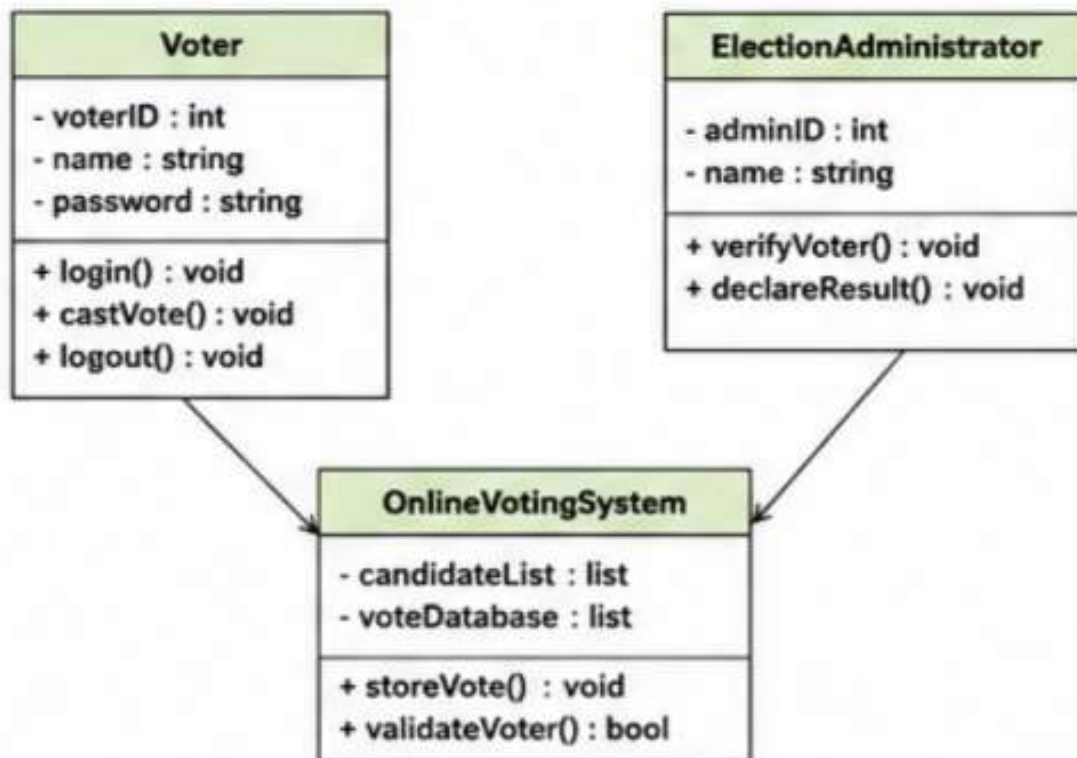
Use case: Voter Registration, Login, Candidate Selection, Vote Casting, Result Declaration, Logout.



CLASS DIAGRAM:

This diagram consists of the following classes, attributes and their operations.

CLASSES	ATTRIBUTES	OPERATIONS
Online voting system	Verify voter details, Store votes	Validate vote ()
Election administrator	Candidate details	Publish result ()
Voter	Voter ID, Name	Cast vote ()

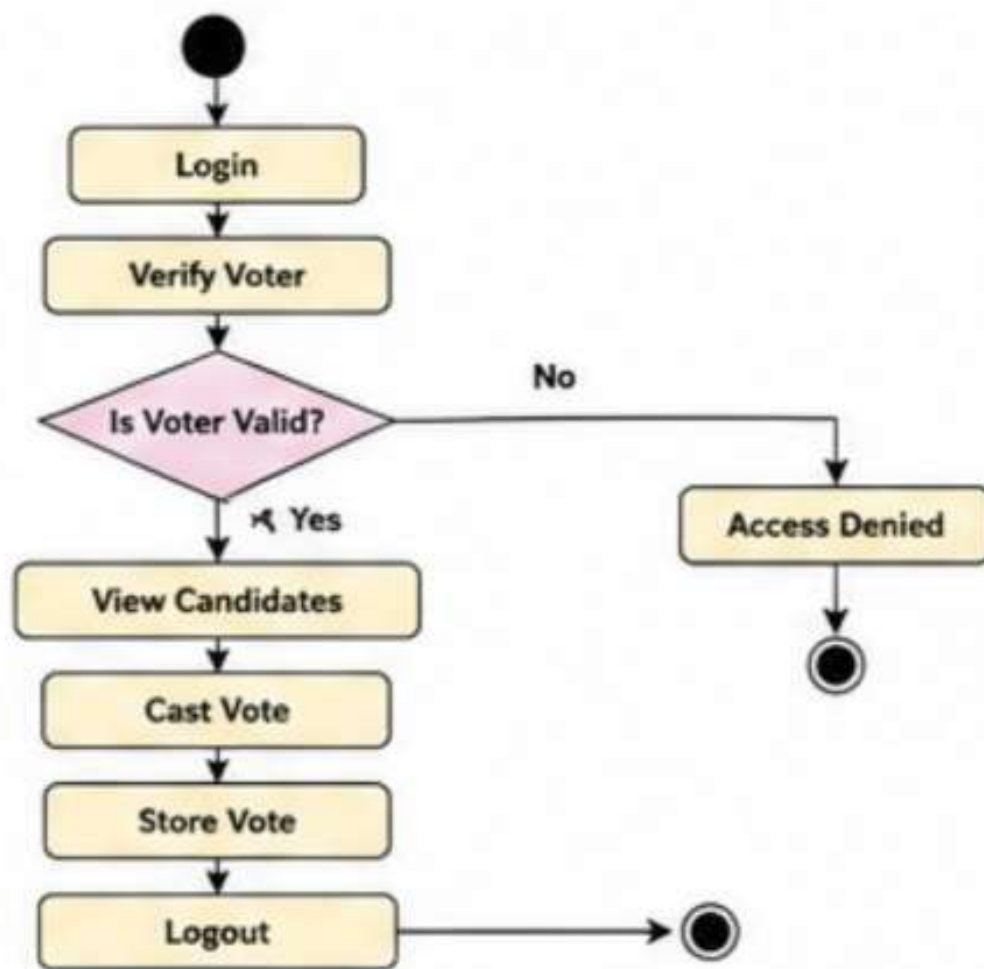


ACTIVITY DIAGRAM:

This diagram will have the activities as Start point, End point, Decision boxes as given below:

Activities: Voter login, Verification of voter, Candidate selection, Vote casting, Result generation.

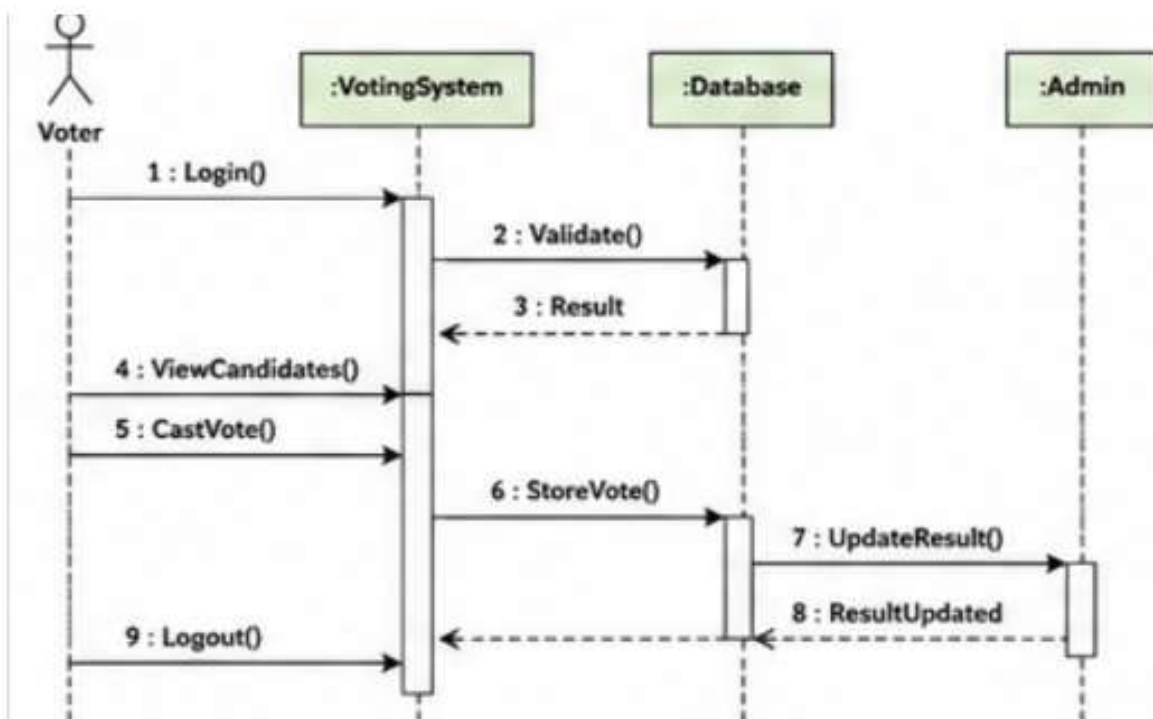
Decision box: Check whether voter is valid or not.



SEQUENCE DIAGRAM:

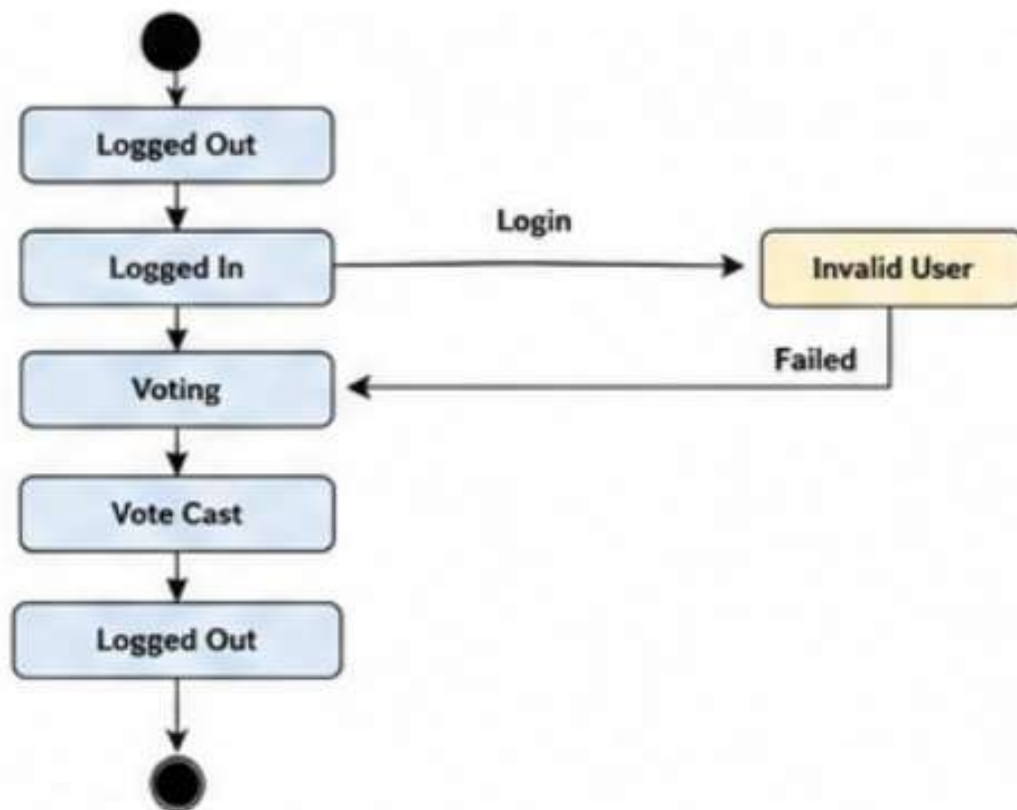
This diagram consists of the objects, messages and return messages.

Object: Voter, Election administrator, Online voting system.



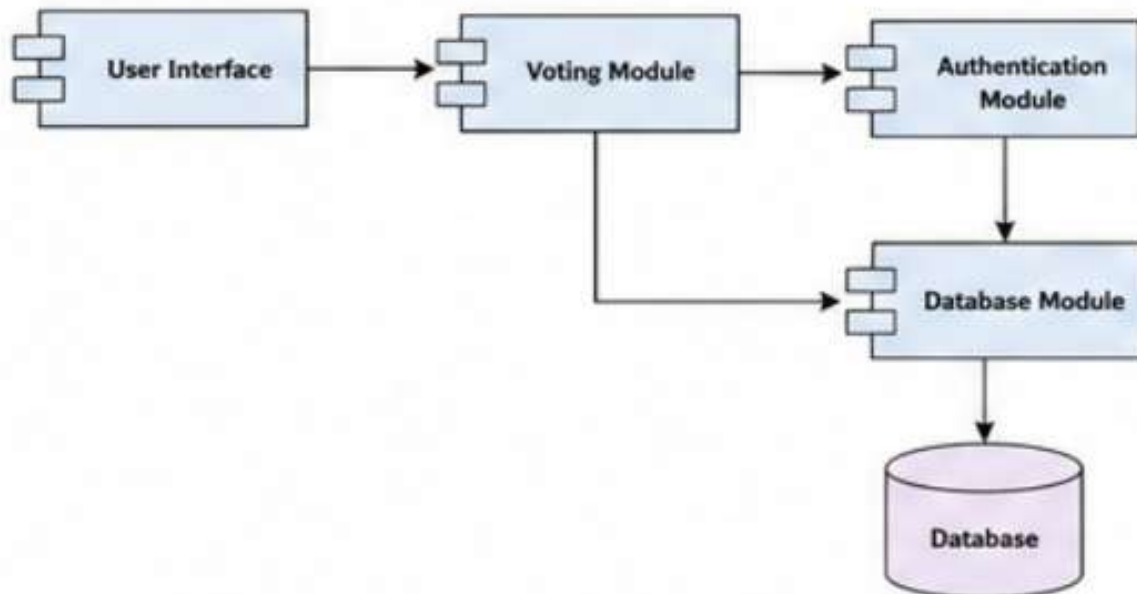
STATE CHART DIAGRAM:

The purpose of the state chart diagram is to represent the different states involved in the Online Voting System during the voting process.



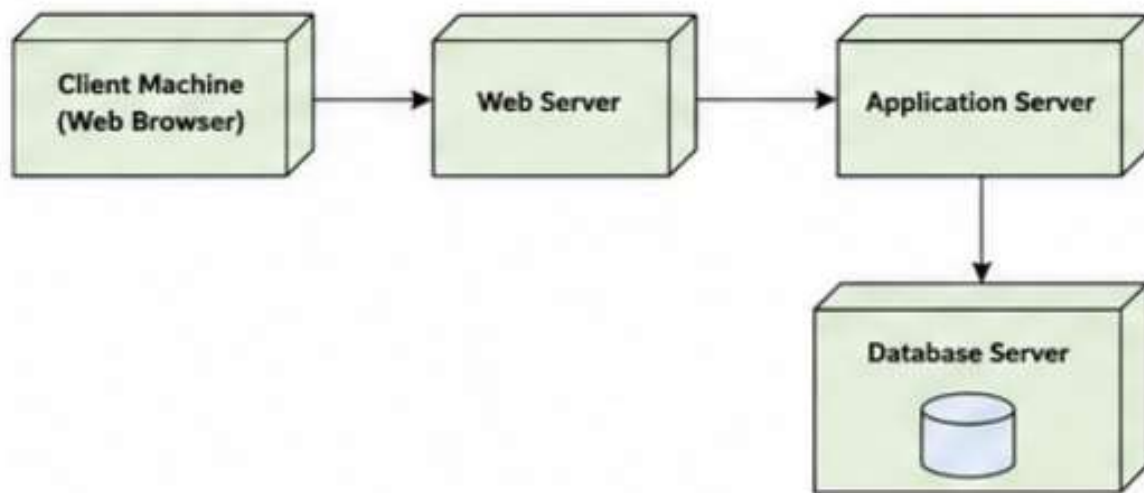
COMPONENT DIAGRAM:

The component diagram's main purpose is to show the structural relationships between the components of the system. It is represented using boxed figures. Dependencies are represented by communication association.



DEPLOYMENT DIAGRAM:

A deployment diagram in UML is used to model the physical deployment of artifacts on nodes. Deployment diagrams show the allocation of software components into hardware nodes. It is represented by 3-dimensional boxes. Dependencies are represented by communication association.



RESULT:

Thus the diagrams [use case, activity, sequence, collaboration, class, component, deployment, package] for the Online Voting System has been designed, executed and output is verified.

EX NO: 4	RAILWAY RESERVATION SYSTEM
DATE:	

AIM:

To draw the diagrams [use case, activity, sequence, collaboration, class, state chart, component, deployment, package] for the Railway Reservation System.

SOFTWARE REQUIREMENTS SPECIFICATION

SL.NO	SOFTWARE REQUIREMENTS SPECIFICATION
1.0	Hardware Requirements
1.1	Software Requirements
1.2	Problem Analysis and Project Plan
1.3	Project Description
1.4	Reference

1.0 HARDWARE REQUIREMENTS:

Intel Pentium Processor i3/i5

1.1 SOFTWARE REQUIREMENTS:

Rational Rose / Argo UML

1.2 PROBLEM ANALYSIS AND PROJECT PLAN

To simplify railway ticket booking, software has been created using Rational Rose tool. Initially, the passenger logs into the railway reservation system and enters journey details. The information is stored in the database and ticket availability is checked. After successful payment and confirmation, the ticket is issued to the passenger.

1.3 PROJECT DESCRIPTION:

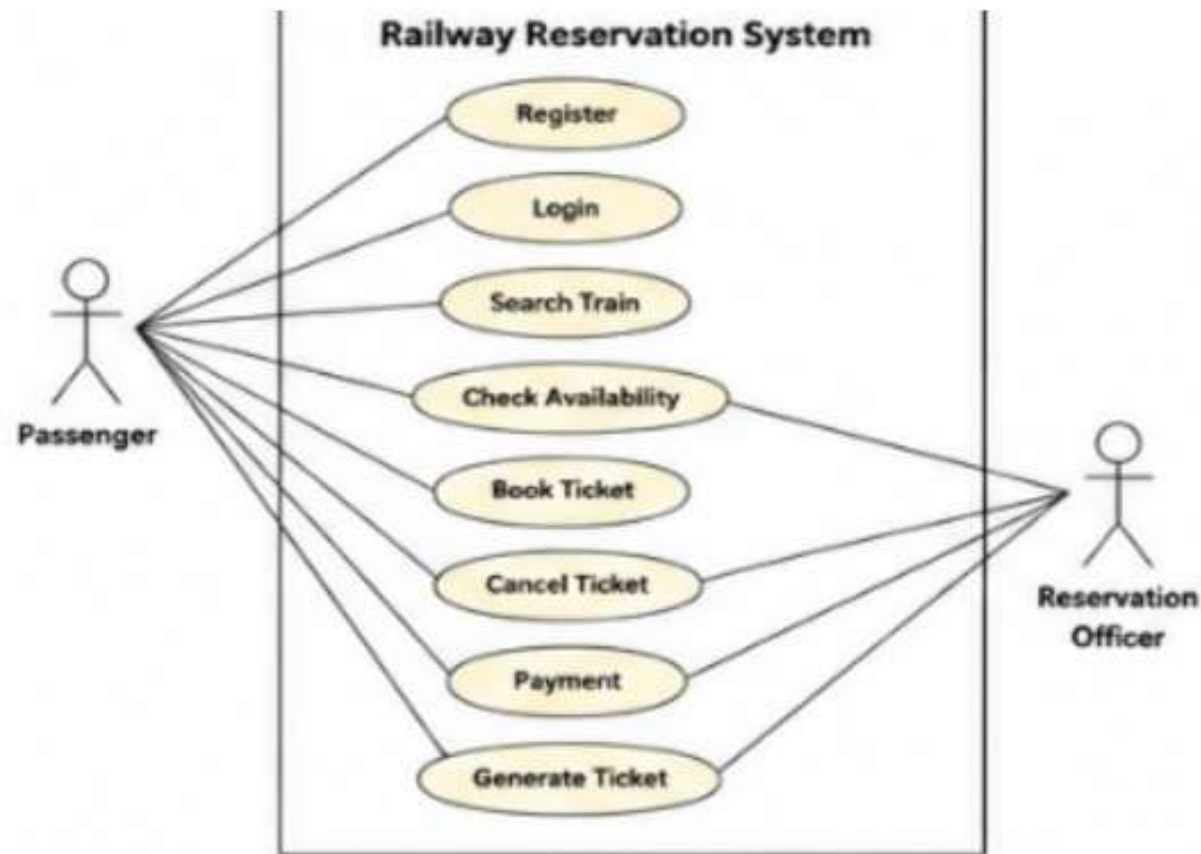
This software is designed for reservation and cancellation of railway tickets through a centralized system. Passenger details and train schedules are verified by the reservation server. The system provides ticket booking, seat allocation, payment processing, and ticket cancellation facilities efficiently.

1.4 REFERENCES:

IEEE Software Requirement Specification format.

USE CASE DIAGRAM:

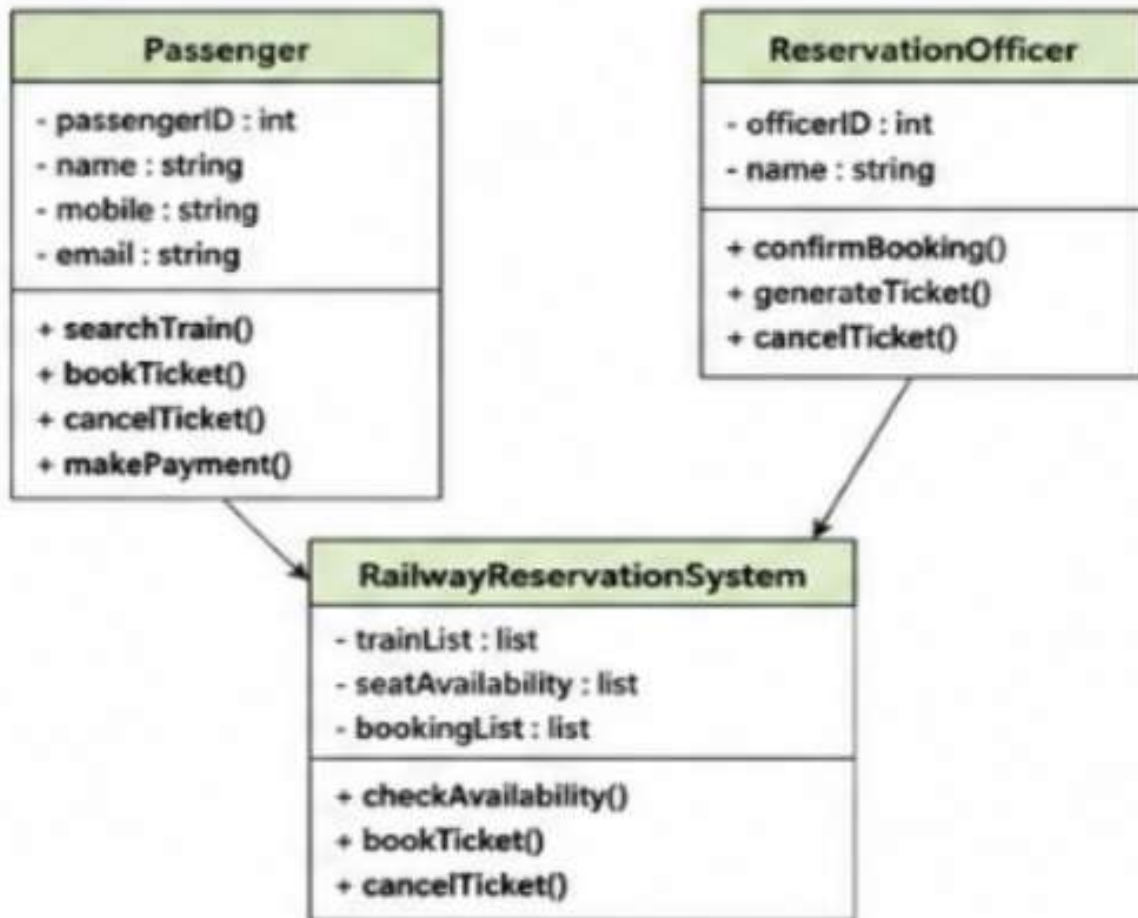
This diagram will contain the actors and use cases which are given below.
Actors: Passenger, Reservation Officer.
Use case: Passenger Registration, Ticket Booking, Seat Availability Check, Ticket Cancellation, Payment, Ticket Confirmation.



CLASS DIAGRAM:

This diagram consists of the following classes, attributes and their operations.

CLASSES	ATTRIBUTES	OPERATIONS
Railway reservation system	Train details, Seat details	Check availability ()
Reservation officer	Passenger details	Confirm booking ()
Passenger	Name, Passenger ID	Book ticket ()

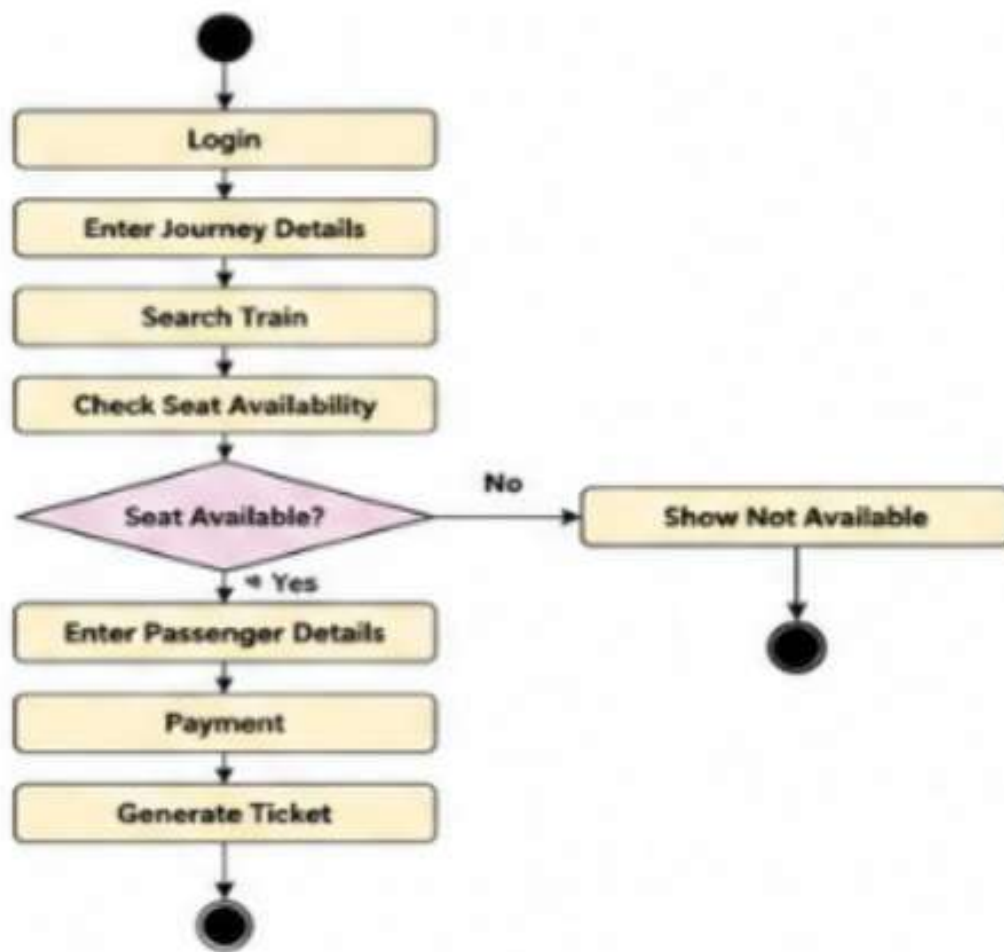


ACTIVITY DIAGRAM:

This diagram will have the activities as Start point, End point, Decision boxes as given below:

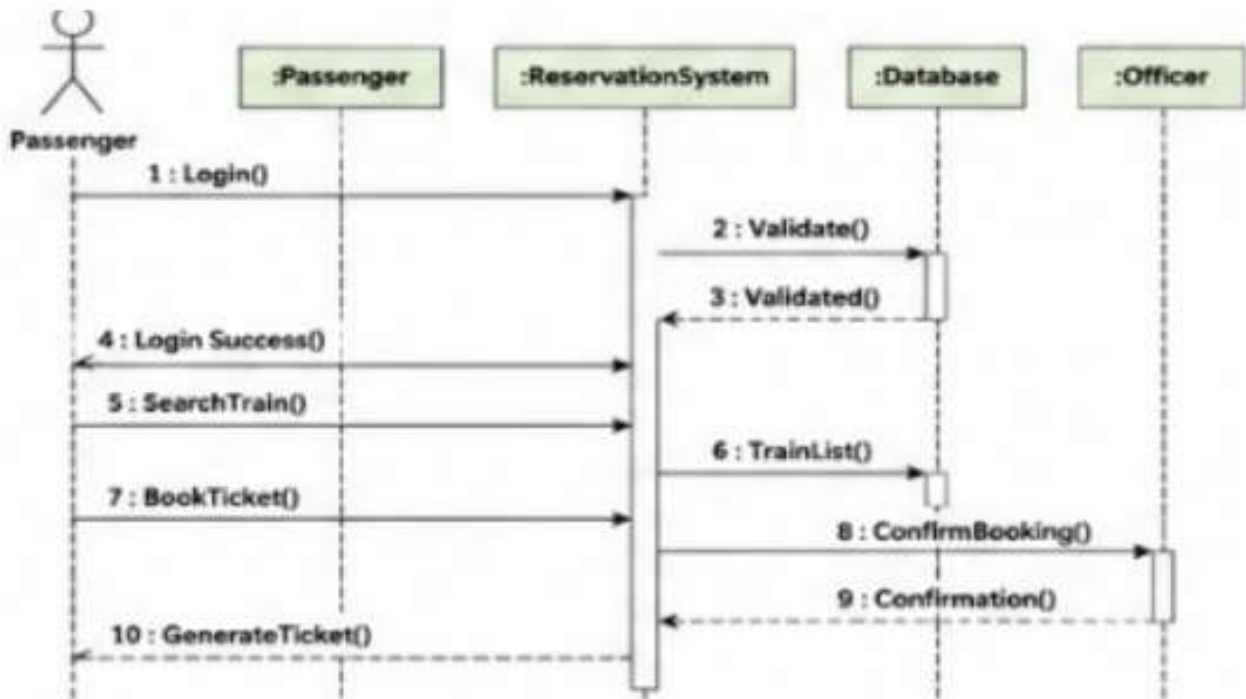
Activities: Login, Enter journey details, Seat checking, Payment process, Ticket confirmation.

Decision box: Check whether seats are available or not.



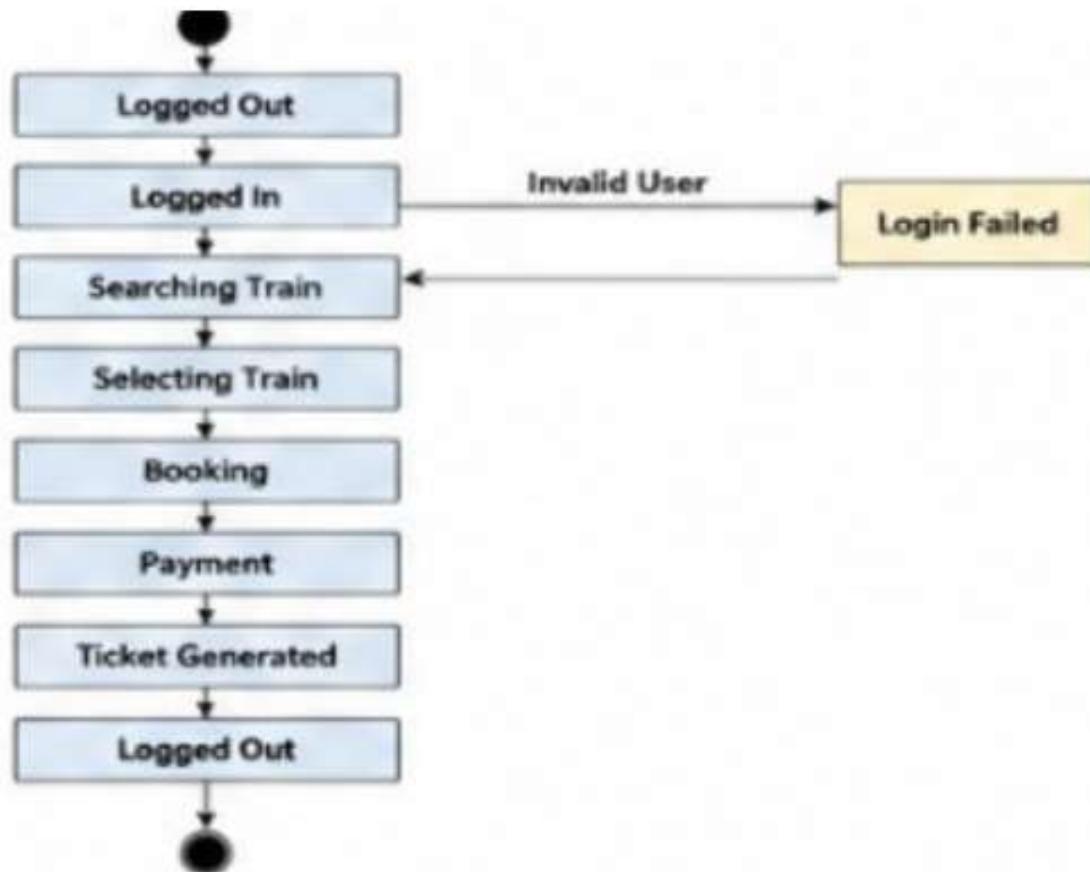
SEQUENCE DIAGRAM:

This diagram consists of the objects, messages and return messages.
Object: Passenger, Reservation officer, Railway reservation system.



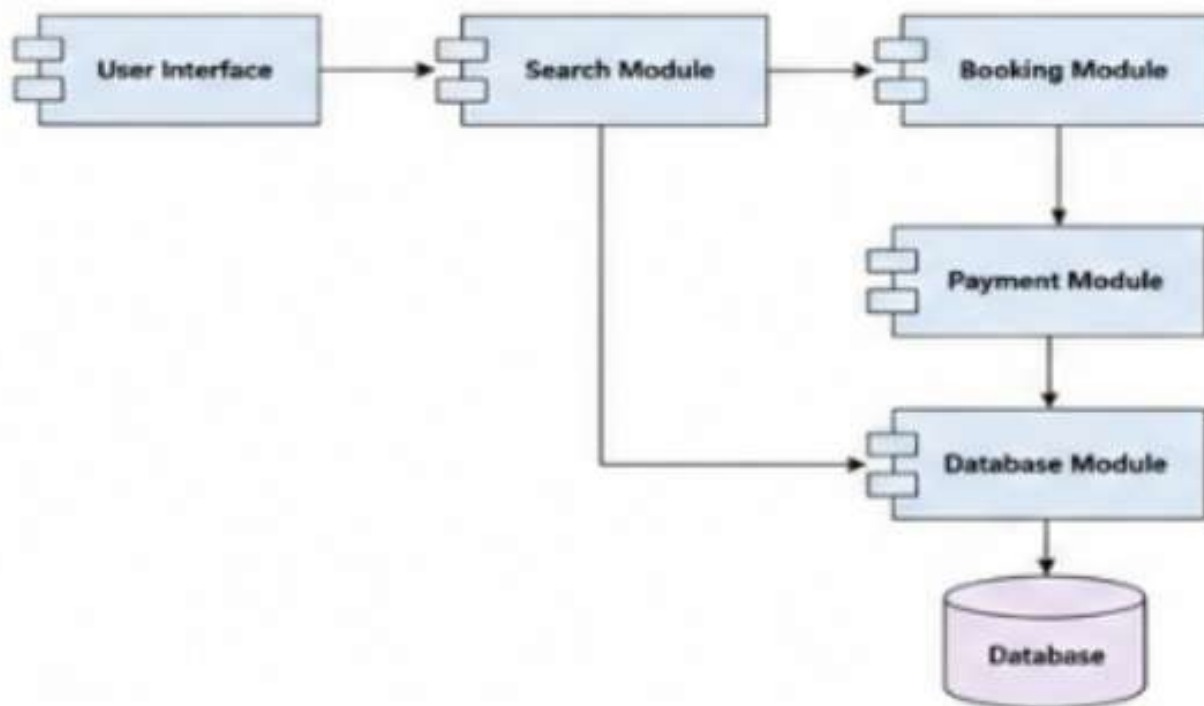
STATE CHART DIAGRAM:

The purpose of the state chart diagram is to represent the various states involved in the Railway Reservation System during ticket booking and cancellation.



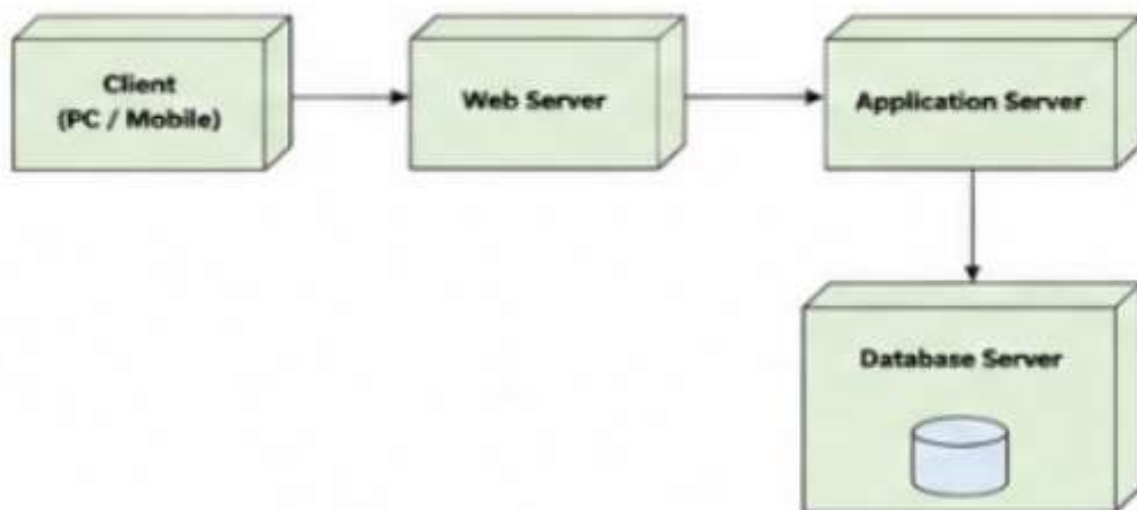
COMPONENT DIAGRAM:

The component diagram's main purpose is to show the structural relationships between the components of a system. It is represented by boxed figures. Dependencies are represented by communication association.



DEPLOYMENT DIAGRAM:

A deployment diagram in UML is used to model the physical deployment of artifacts on deployment targets. Deployment diagrams show the allocation of software components into hardware nodes. It is represented by 3-dimensional boxes. Dependencies are represented by communication association.



RESULT:

Thus the diagrams [use case, activity, sequence, collaboration, class, component, deployment, packa

for the Railway Reservation System has been designed, executed and output is verified.

EX NO: 5	ONLINE SHOPPING SYSTEM
DATE:	

AIM:

To draw the diagrams [use case, activity, sequence, collaboration, class, statechart, component, deployment, package] for the Online Shopping System.

SOFTWARE REQUIREMENTS SPECIFICATION

SL.NO	SOFTWARE REQUIREMENTS SPECIFICATION
1.0	Hardware Requirements
1.1	Software Requirements
1.2	Problem Analysis and Project Plan
1.3	Project Description
1.4	Reference

1.9HARDWARE REQUIREMENTS:

- Intel Pentium Processor I3/I5
- Minimum 4 GB RAM
- Hard Disk: 500 GB

1.10 SOFTWARE REQUIREMENTS:

Operating System: Windows 7/10/11

UML Tool: Rational Rose / Argo UML

1.11 PROBLEM ANALYSIS AND PROJECT PLANNING

The Online Shopping System is designed to simplify the process of purchasing products through online platforms. Customers can search products, add items to cart, and place orders online. The administrator manages product details, payments, and order records. The system ensures secure transactions and maintains customer and product information efficiently.

1.12 PROJECT DESCRIPTION:

This software is developed to maintain the complete details of products and customers in an online shopping system.

The system allows:

- Searching products
- Adding products to cart
- Online payment
- Order confirmation

- Maintaining customer records

The system verifies customer details and updates the database accordingly. This system reduces manual work and improves efficiency.

1.9 REFERENCES:

IEEE Software Requirement Specification format.

USE CASE DIAGRAM:

This diagram will contain the actors, use cases which are given below

Actors:

Customer/User, Administrator

Use case:

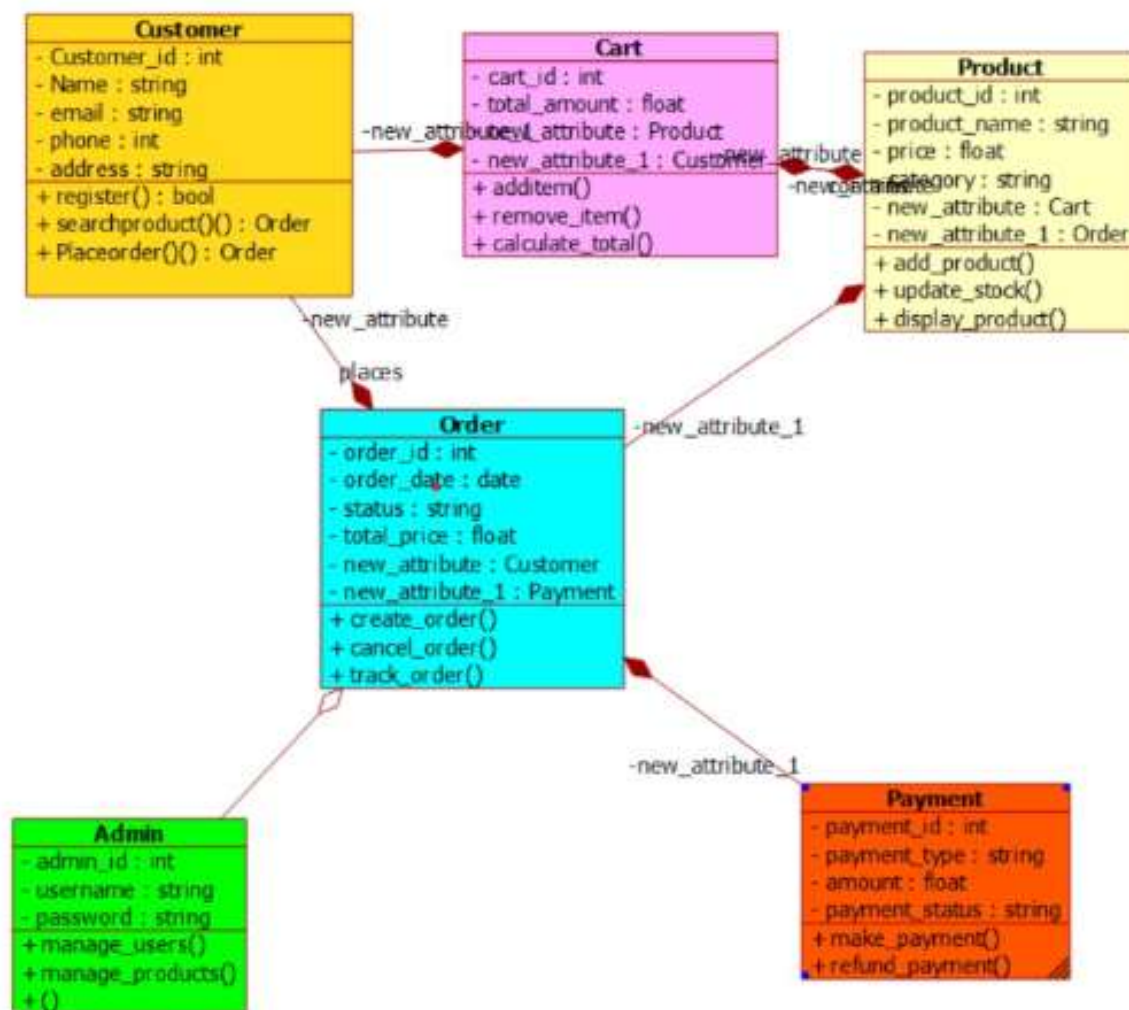
Search product, view product details, add to cart, place order, make payment, cancel order, maintain records



CLASS DIAGRAM:

This diagram consists of the following classes, attributes and their operations

CLASSES	ATTRIBUTES	OPERATIONS
Online Shopping System	Product and customer details	Manage orders(), Maintain records()
User (Customer)	Name, ID, Cart details	Search product(), Place order(), Cancel order()
Administrator	Product records, order records	Add product(), Update records()



ACTIVITY DIAGRAM:

This diagram will have the activities as Start point, End point, Decision boxes as given below:

Activities:

Enter customer details, search products, add to cart, payment process, order confirmation

Decision box:

Check the payment status



STATE CHART DIAGRAM (Restaurant Management System):

The purpose of a state chart diagram is to understand the process involved in performing a method in the Online Shopping System. It is also called a state diagram. A state is represented as a rounded box, which may contain one or more compartments. An initial state is represented as a small dot, and a final state is represented as a circle surrounding a small dot.

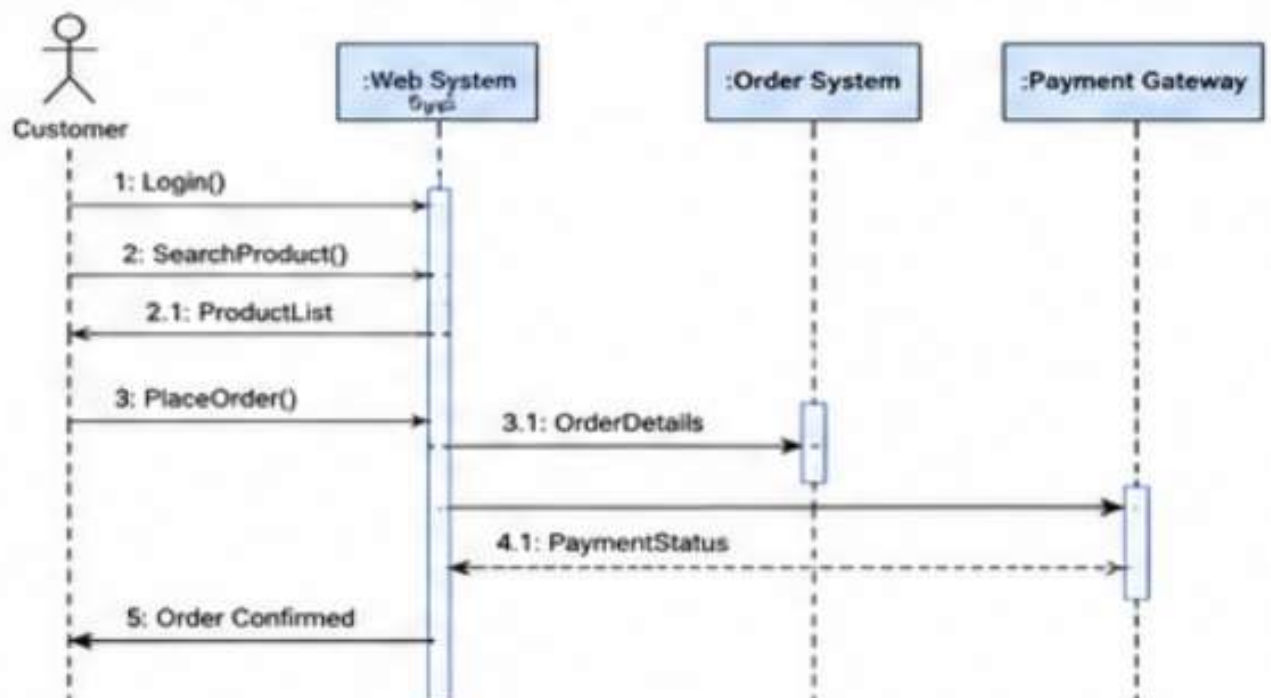


SEQUENCE DIAGRAM (Online shopping System):

This diagram consists of the objects, messages, and return messages.

Objects:

Customer/User, Administrator, Online Shopping System

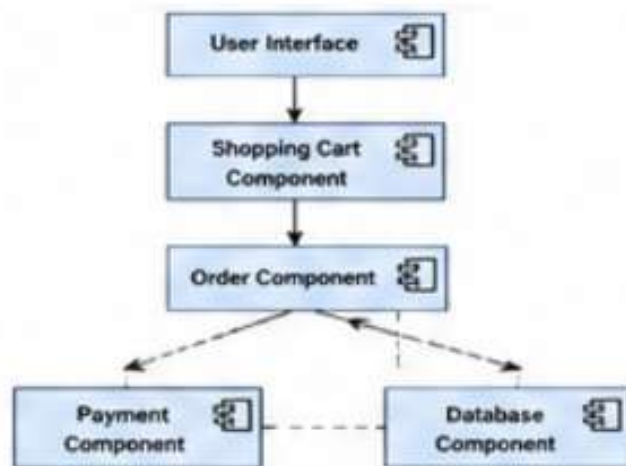


COMPONENT DIAGRAM:

The main purpose of the component diagram is to show the structural relationships between the components of the Online Shopping System. It is represented by boxed figures, and dependencies are represented by communication associations.

Components:

User Interface, Product Management Module, Payment Module, Order Management Module, Database

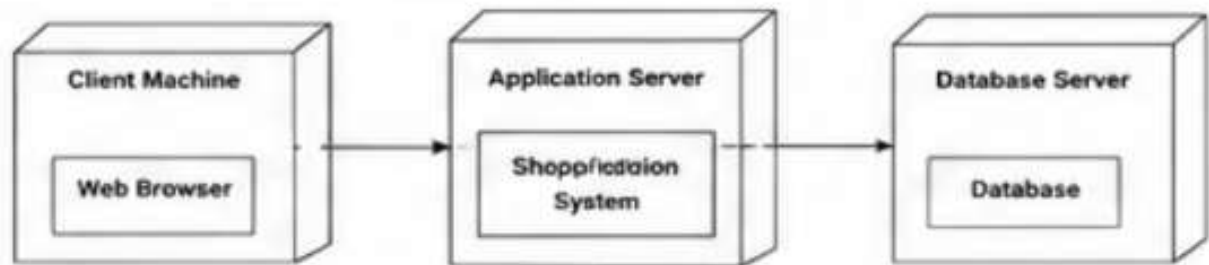


DEPLOYMENT DIAGRAM:

A deployment diagram in UML is used to model the physical deployment of artifacts on nodes. Deployment diagrams show the allocation of software components into hardware nodes. It is represented by 3-dimensional boxes. Dependencies are represented by communication association.

Nodes:

- Customer/Admin System
- Application Server
- Database Server



RESULT:

Thus the diagrams [use case, activity, sequence, collaboration, class, statechart, component, deployment package] for the Online Shopping System have been designed and verified successfully.

EX NO: 6	ATM MANAGEMENT SYSTEM
DATE:	

AIM:

To draw the diagrams [use case, activity, sequence, collaboration, class, statechart, component, deployment, package] for the ATM Management System.

SOFTWARE REQUIREMENTS SPECIFICATION

SL.NO	SOFTWARE REQUIREMENTS SPECIFICATION
1.0	Hardware Requirements
1.1	Software Requirements
1.2	Problem Analysis and Project Plan
1.3	Project Description
1.4	Reference

1.13 HARDWARE REQUIREMENTS:

- Intel Pentium Processor I3/I5
- Minimum 4 GB RAM
- Hard Disk: 500 GB

1.14 SOFTWARE REQUIREMENTS:

Operating System: Windows 7/10/11

UML Tool: Rational Rose / Argo UML

1.15 PROBLEM ANALYSIS AND PROJECT PLANNING

The ATM Management System is designed to simplify banking transactions through ATM machines. Customers can withdraw cash, deposit money, check balance, and transfer funds securely. The bank administrator manages

account details and transaction records. The system ensures secure authentication and fast transaction processing.

1.16 PROJECT DESCRIPTION:

This software is developed to maintain the complete details of customer accounts and ATM transactions.

The system allows:

- Cash withdrawal
- Cash deposit
- Balance enquiry
- Fund transfer
- Maintaining transaction records

The ATM system verifies customer details and updates the database accordingly. This system reduces manual banking work and improves efficiency.

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1.17 REFERENCES:

IEEE Software Requirement Specification format

USE CASE DIAGRAM:

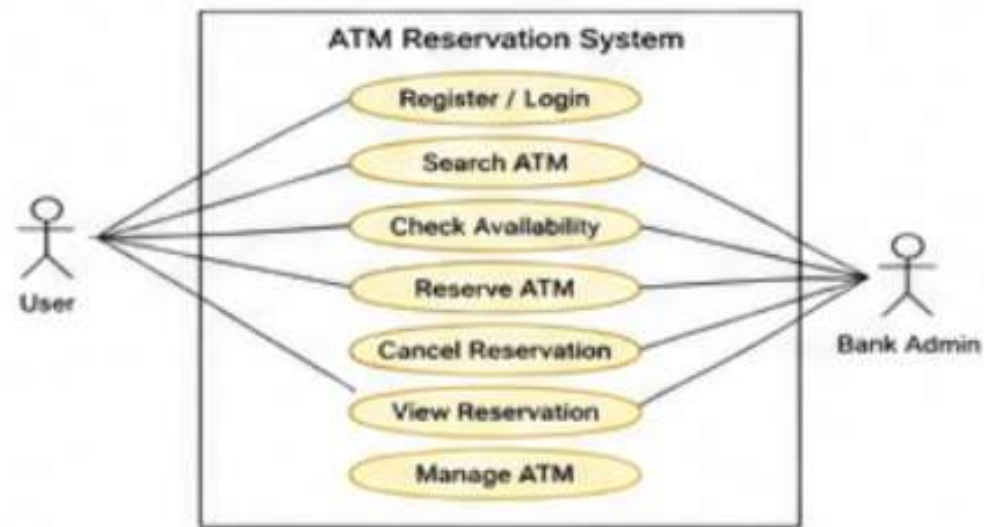
This diagram will contain the actors, use cases which are given below

Actors:

Customer/User, Bank Administrator

Use case:

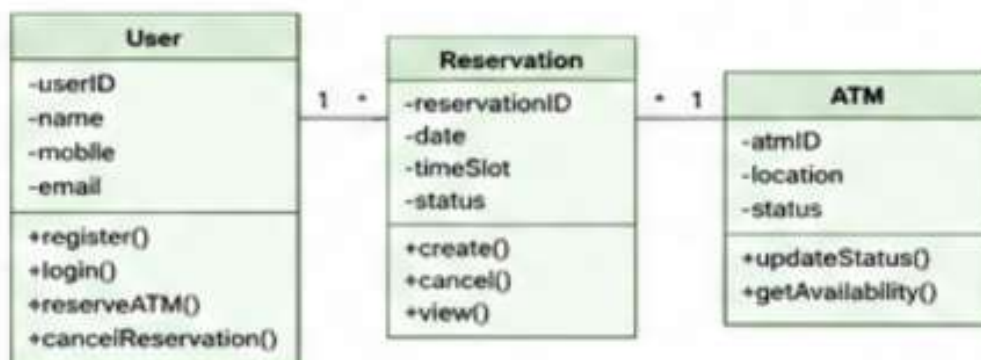
Insert card, enter PIN, withdraw cash, deposit money, check balance, transfer funds, maintain records



CLASS DIAGRAM:

This diagram consists of the following classes, attributes and their operations

CLASSES	ATTRIBUTES	OPERATIONS
ATM System	Account and transaction details	Verify PIN(), Maintain records()
User (Customer)	Name, Account number	Withdraw cash(), Check balance()
Bank Administrator	Customer records	Update account(), Generate report()

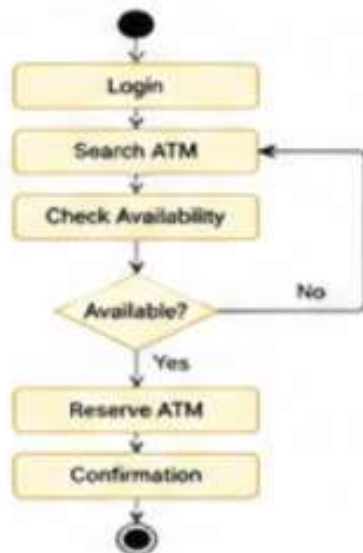


ACTIVITY DIAGRAM:

This diagram will have the activities as Start point, End point, Decision boxes as given below:

Activities:

Insert ATM card, enter PIN, select transaction, process transaction, print receipt

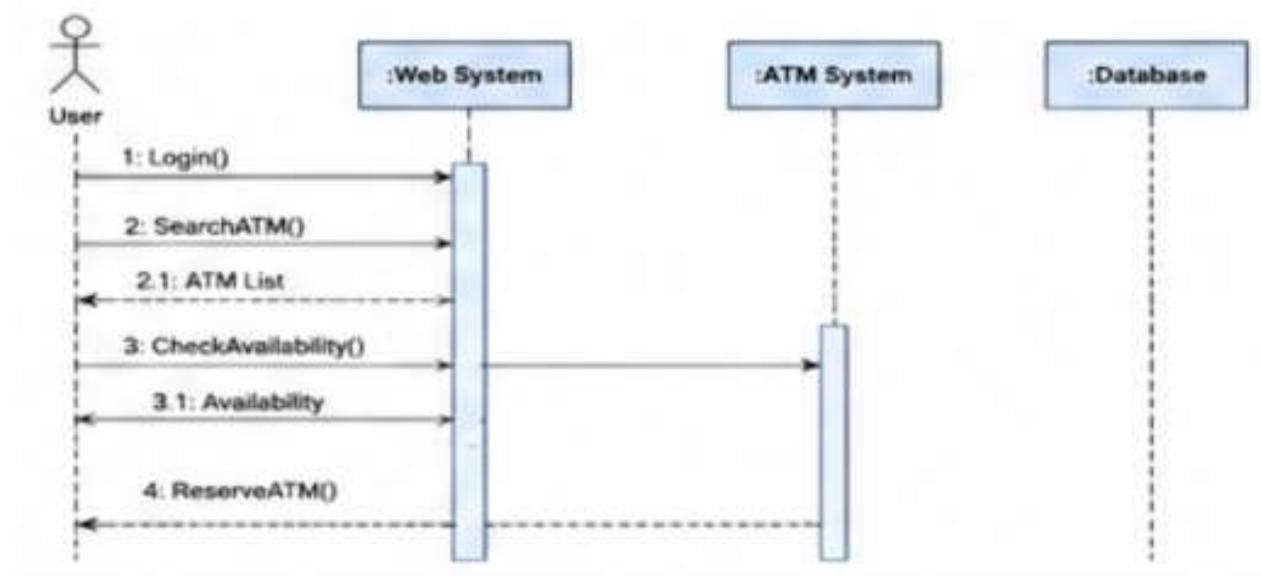


SEQUENCE DIAGRAM:

This diagram consists of the objects, messages, and return messages.

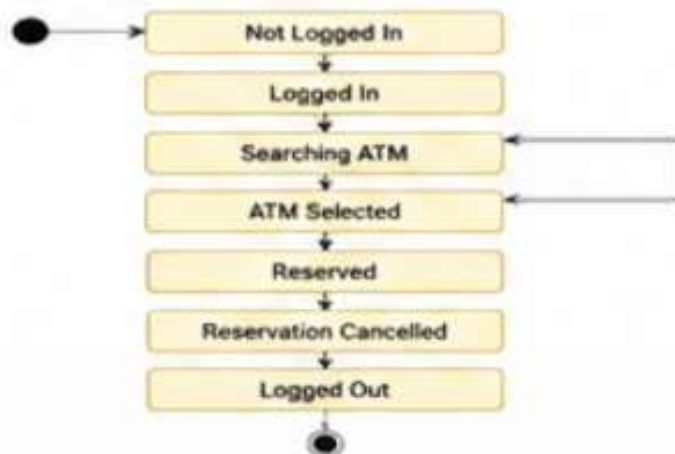
Objects:

Customer/User, ATM System, Bank Server



STATE CHART DIAGRAM:

The purpose of a state chart diagram is to understand the process involved in performing a method in the ATM Management System. It is also called a state diagram. A state is represented as a rounded box, which may contain one or more compartments. An initial state is represented as a small dot, and a final state is represented as a circle surrounding a small dot.

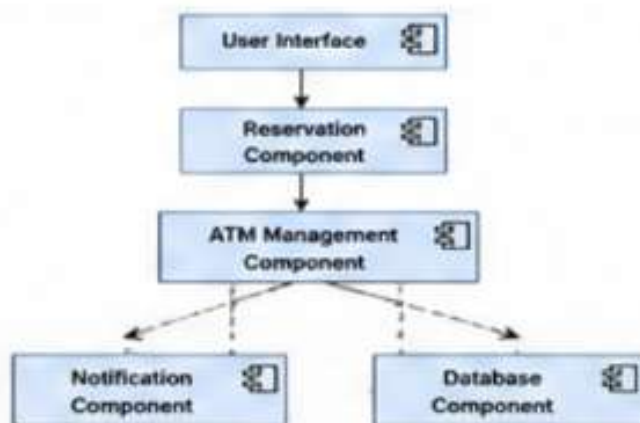


COMPONENT DIAGRAM:

The main purpose of the component diagram is to show the structural relationships between the components of the ATM Management System. It is represented by boxed figures, and dependencies are represented by communication associations.

Components:

User Interface, Authentication Module, Transaction Module, Account Management Module, Database



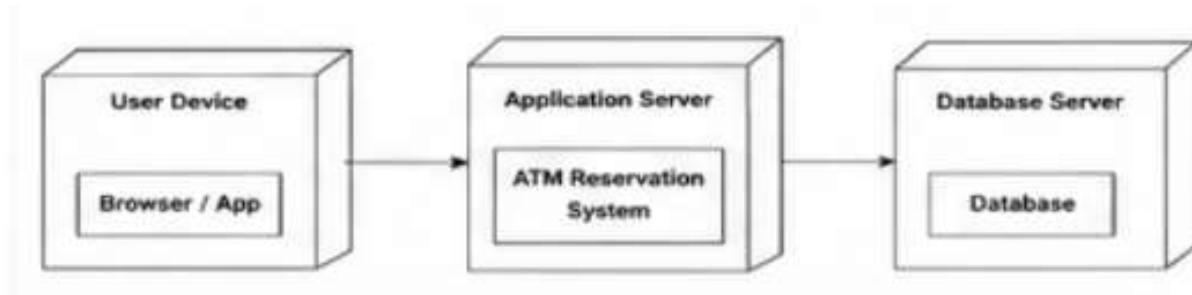
DEPLOYMENT DIAGRAM:

A deployment diagram in UML is used to model the physical deployment of artifacts on nodes. Deployment diagrams show the allocation of software components into hardware nodes. It is represented by 3-dimensional boxes. Dependencies are represented by communication

association.

Nodes:

Client System, Application Server, Database Server



RESULT:

Thus the diagrams [use case, activity, sequence, collaboration, class, statechart, component, deployment, package] for the ATM Management System have been designed and verified successfully.

Ex No:7	PASSPORT AUTOMATION SYSTEM
Date:	

AIM:

To draw the diagrams [usecase, activity, sequence, collaboration, class, statechart, collaboration, component, deployment, package] for the Passport Automation System.

SOFTWARE REQUIREMENTS SPECIFICATION

SL.NO	SOFTWARE REQUIREMENTS SPECIFICATION
1.0	Hardware Requirements
1.1	Software Requirements
1.2	Problem Analysis and Project Plan
1.3	Project Description
1.4	Reference

1.0 HARDWARE REQUIREMENTS:

Intel Pentium Processor I3/I5

1.1 SOFTWARE REQUIREMENTS:

Rational rose /Argo UML

1.2 PROBLEM ANALYSIS AND PROJECT PLAN

To simplify the process of applying passport, software has been created by designing through rational rose tool. Initially the applicant login the passport automation system and submits his details. These details are stored in the database and verification process done by the passport administrator, regional administrator and police the passport is issued to the applicant.

1.3 PROJECT DESCRIPTION:

This software is designed for the verification of the passport details of the applicant by the central computer. The details regarding the passport will be provided to the central computer and the computer will verify the details of applicant and provide approval to the office. Then the passport will issue from the office to the applicant.

1.4 REFERENCES:

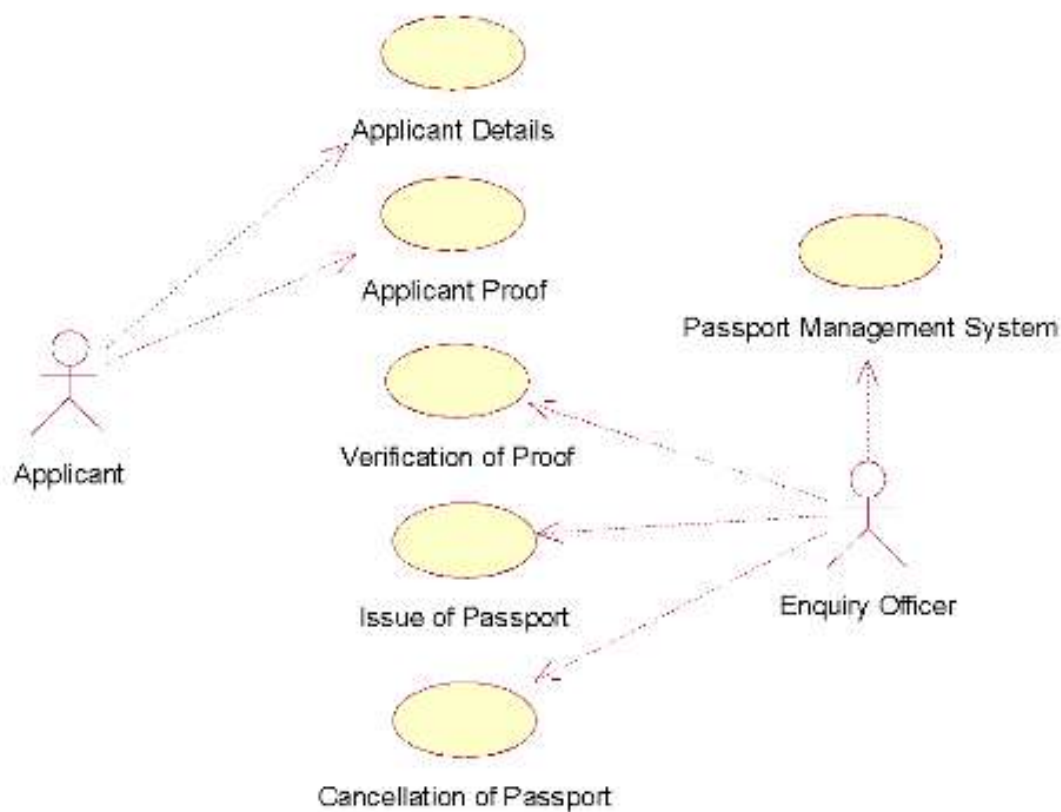
IEEE Software Requirement Specification format.

USE CASE DIAGRAM:

This diagram will contain the actors, use cases which are given below

Actors: Applicant, Enquiry Officer.

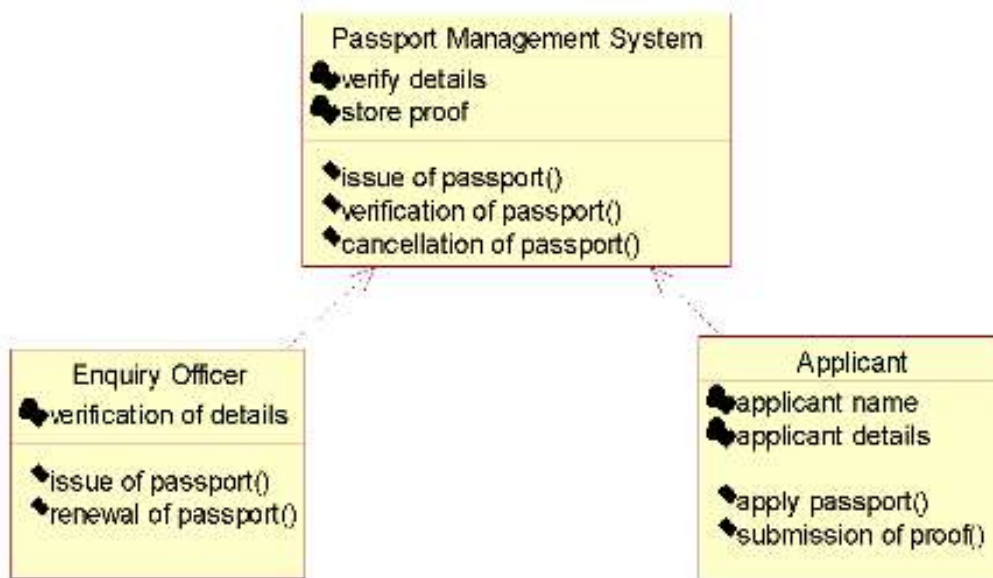
Use case: Applicant details, Applicant proof, Verification of proof, Issue of passport, Cancellation of the passport.



CLASS DIAGRAM:

This diagram consists of the following classes, attributes and their operations.

CLASSES	ATTRIBTES	OPERATIONS
Passport management system	Verify details, Store proof	Verification of proof()
Enquiry officer	Applicant details	Issue of passport()
Applicant	Name, Details	Apply passport()



ACTIVITY DIAGRAM:

This diagram will have the activities as Start point, End point, Decision boxes as given below:

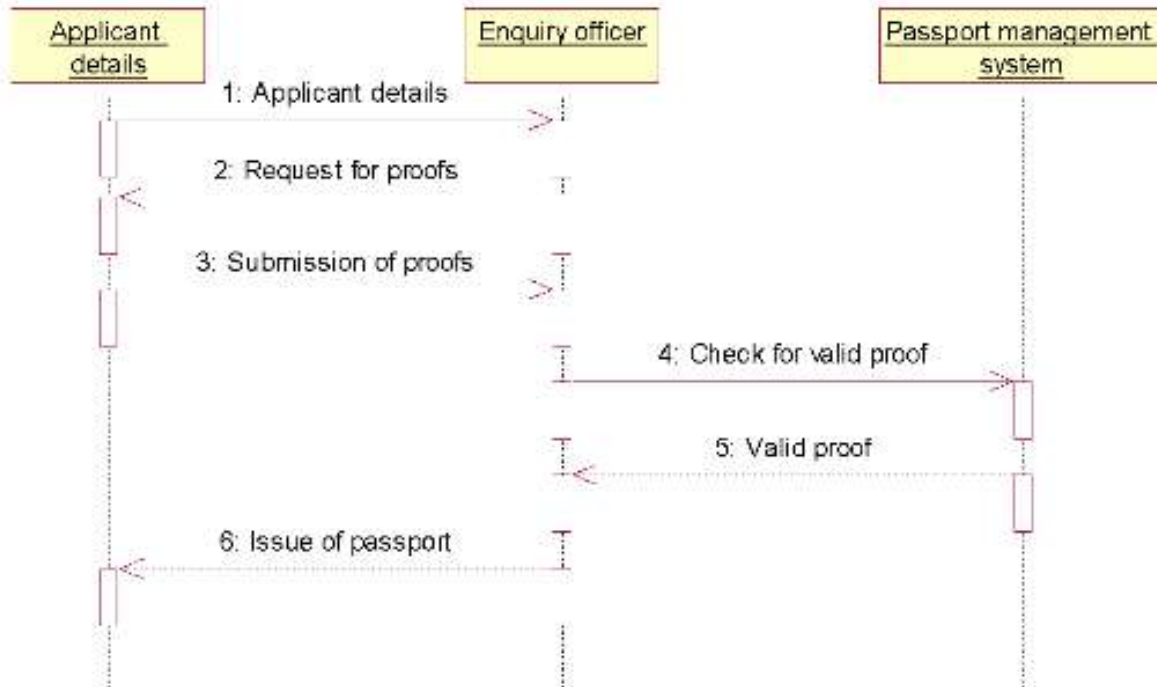
Activities: Enter applicant details, Submission of proof, Verification of details, Issue of passport.

Decision box: Check details whether it is correct or not.

SEQUENCE DIAGRAM:

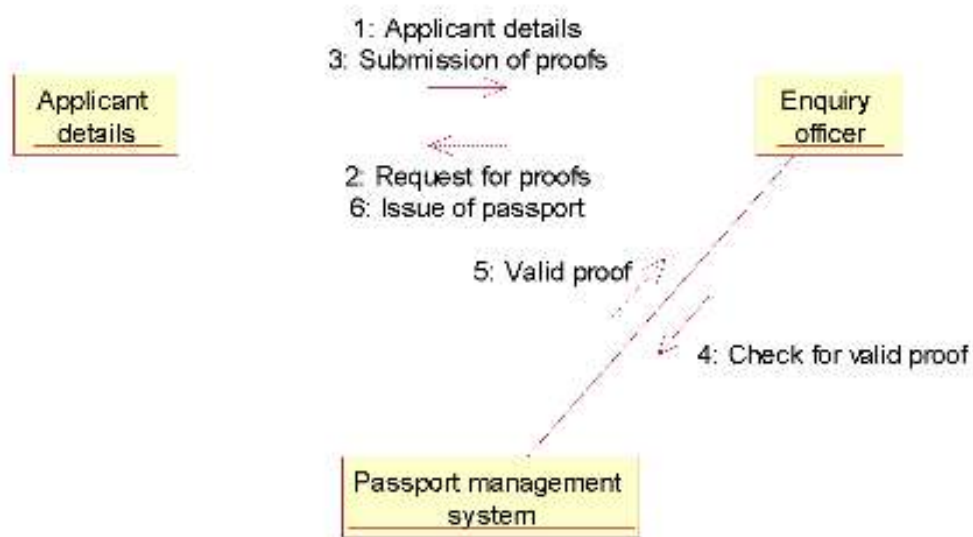
This diagram consists of the objects, messages and return messages.

Object: Applicant, Enquiry officer, Passport management system.



COLLABORATION DIAGRAM:

This diagram contains the objects and actors. This will be obtained by the completion of the sequence diagram and pressing the F5 key.



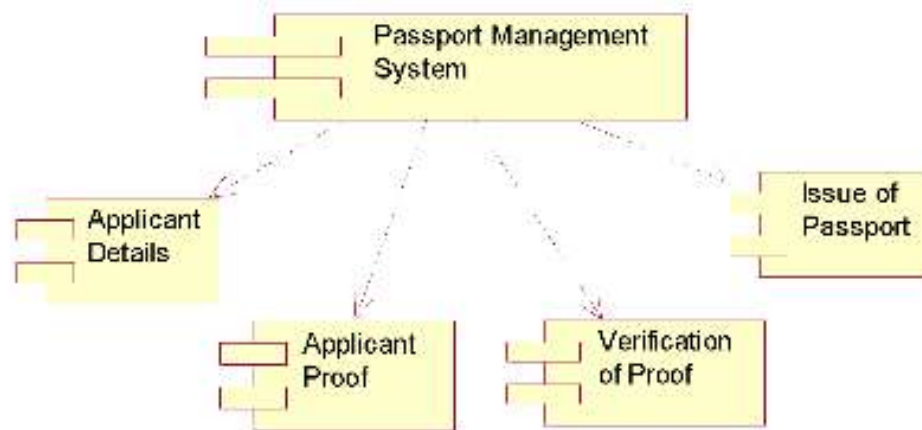
STATE CHART DIAGRAM :

The purpose of state chart diagram is to understand the algorithm involved in performing a method. It is also called as state diagram. A state is represented as a round box, which may contain one or more compartments. An initial state is represented as small dot. An final state is represented as circle surrounding a small dot.



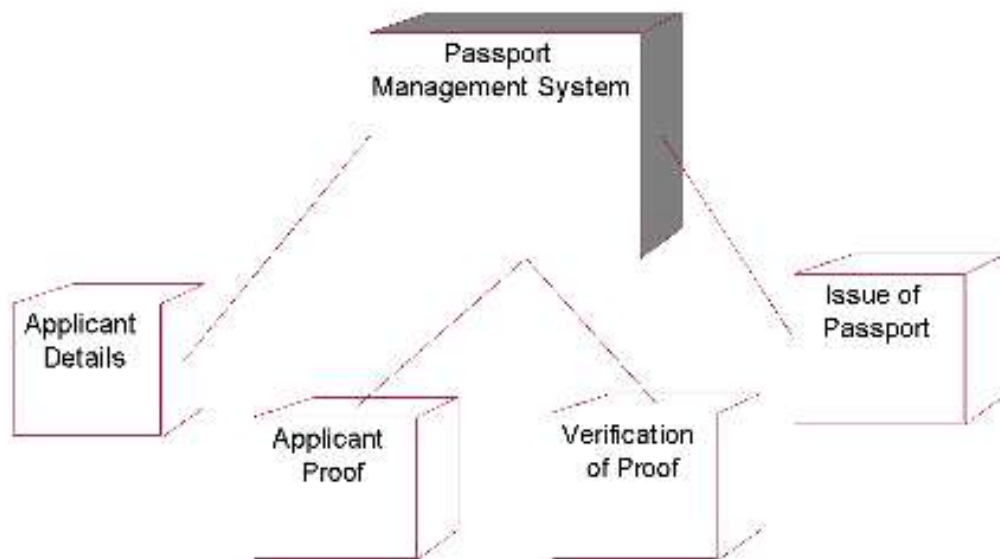
COMPONENT DIAGRAM

The component diagram's main purpose is to show the structural relationships between the components of a system. It is represented by boxed figure. Dependencies are represented by communication association.



DEPLOYMENT DIAGRAM

A deployment diagram in the unified modeling language serves to model the physical deployment of artifacts on deployment targets. Deployment diagrams show "the allocation of artifacts to nodes according to the Deployments defined between them. It is represented by 3- dimensional box. Dependencies are represented by communication association.



RESULT:

Thus the diagrams [use case, activity, sequence, collaboration, class, collaboration, component, deployment, package] for the Passport Automation system has been designed, executed and output is verified.

Ex.No:8	BOOK BANK REGISTRATION SYSTEM
Date:	

AIM:

To draw the diagrams [usecase, activity, sequence, collaboration, class, statechart, collaboration, component, deployment, package] for the Book bank registration system.

SOFTWARE REQUIREMENTS SPECIFICATION

SL.NO	SOFTWARE REQUIREMENTS SPECIFICATION
1.0	Hardware Requirements
1.1	Software Requirements
1.2	Problem Analysis and Project Plan
1.3	Project Description
1.4	Reference

1.0 HARDWARE REQUIREMENTS:

Intel Pentium Processor I3/I5

1.1 SOFTWARE REQUIREMENTS:

Rational rose /Argo UML

1.2 PROBLEM ANALYSIS AND PROJECT PLAN

To simplify the process of applying passport, software has been created by designing through rational rose tool. Initially the applicant login the passport automation system and submits his details. These details are stored in the database and verification process done by the passport administrator, regional administrator and police the passport is issued to the applicant.

1.3 PROJECT DESCRIPTION:

This software is designed for the verification of the details of the student by the central computer. The details regarding the student will be provided to the central computer through the administrator in the book bank and the computer will verify the details of student and provide approval to the office. Then the books that are needed by the student

will issue from the office to the him.

1.4 REFERENCES:

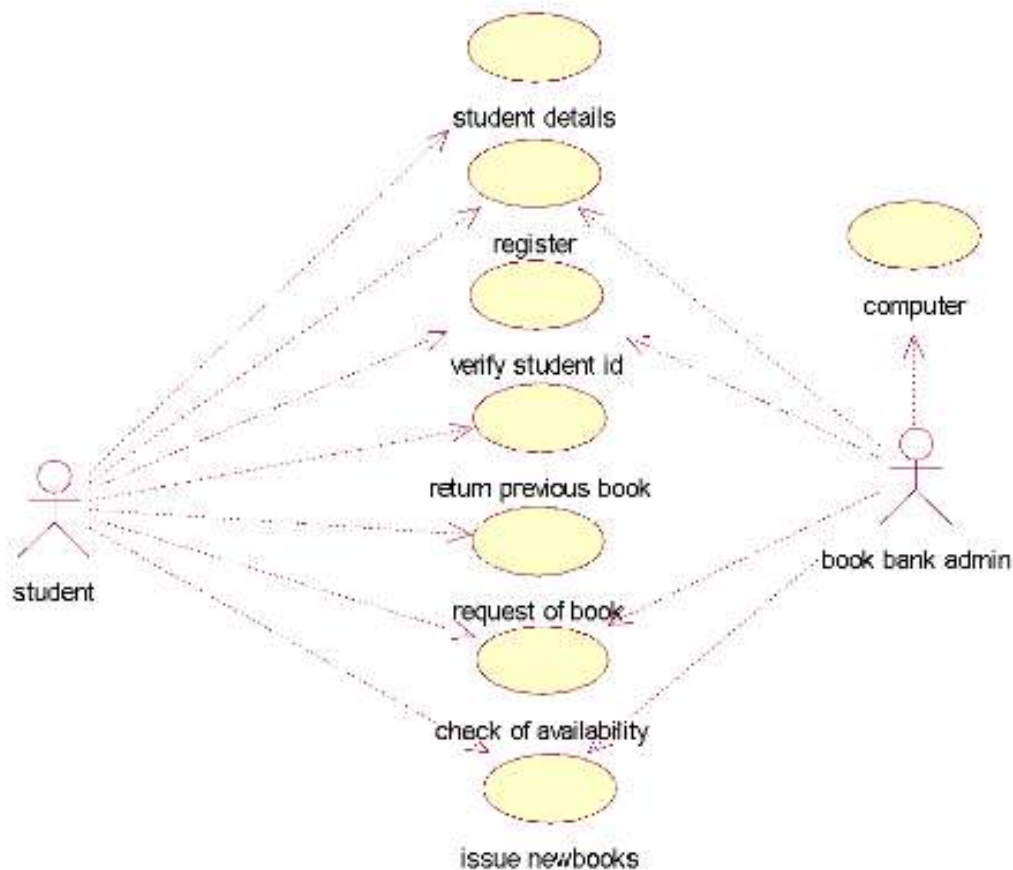
IEEE Software Requirement Specification format.

USE CASE DIAGRAM:

This diagram will contain the actors, use cases which are given below

Actors: Student, book bank admin.

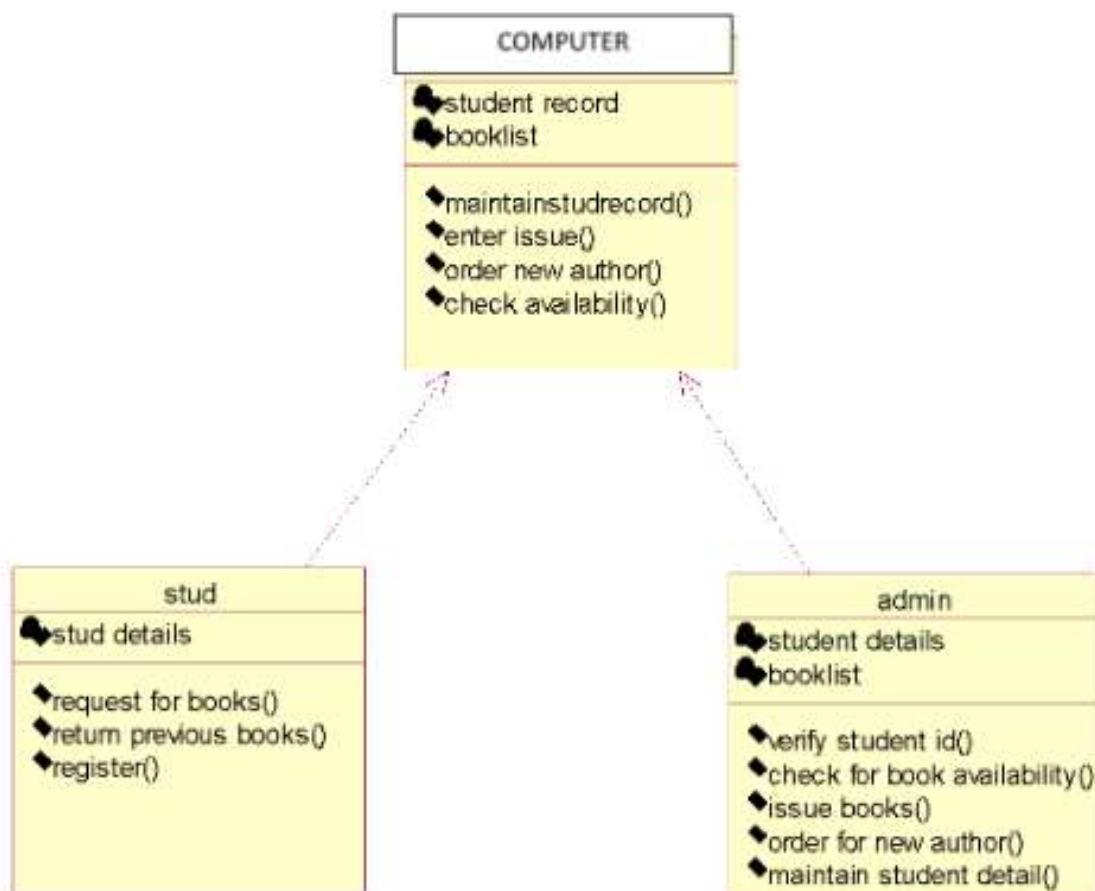
Use case: Student details, register, verify student id, return previous books, request of books, issue of books, check of book availability.



CLASS DIAGRAM:

This diagram consists of the following classes, attributes and their operations.

CLASSES	ATTRIBTES	OPERATIONS
Computer	Student record, Book list	Enter issue(), Check availability()
Stud	Student details	Request for books(), Register()
Admin	Student details, Book list	Verify student id(), Issue books()

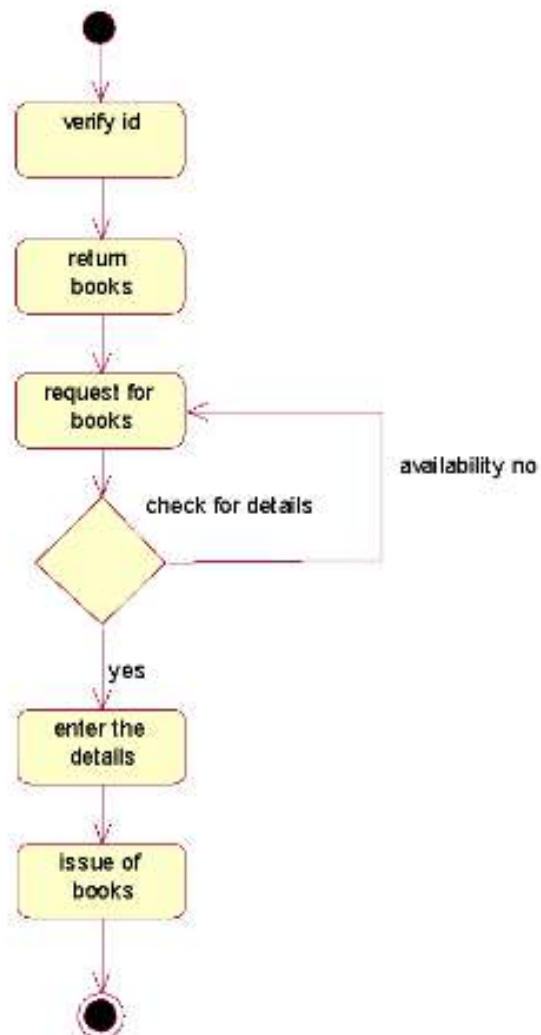


ACTIVITY DIAGRAM:

This diagram will have the activities as Start point, End point, Decision boxes as given below:

Activities: Verify id, return books, request for books, enter book issue details in system, issue books

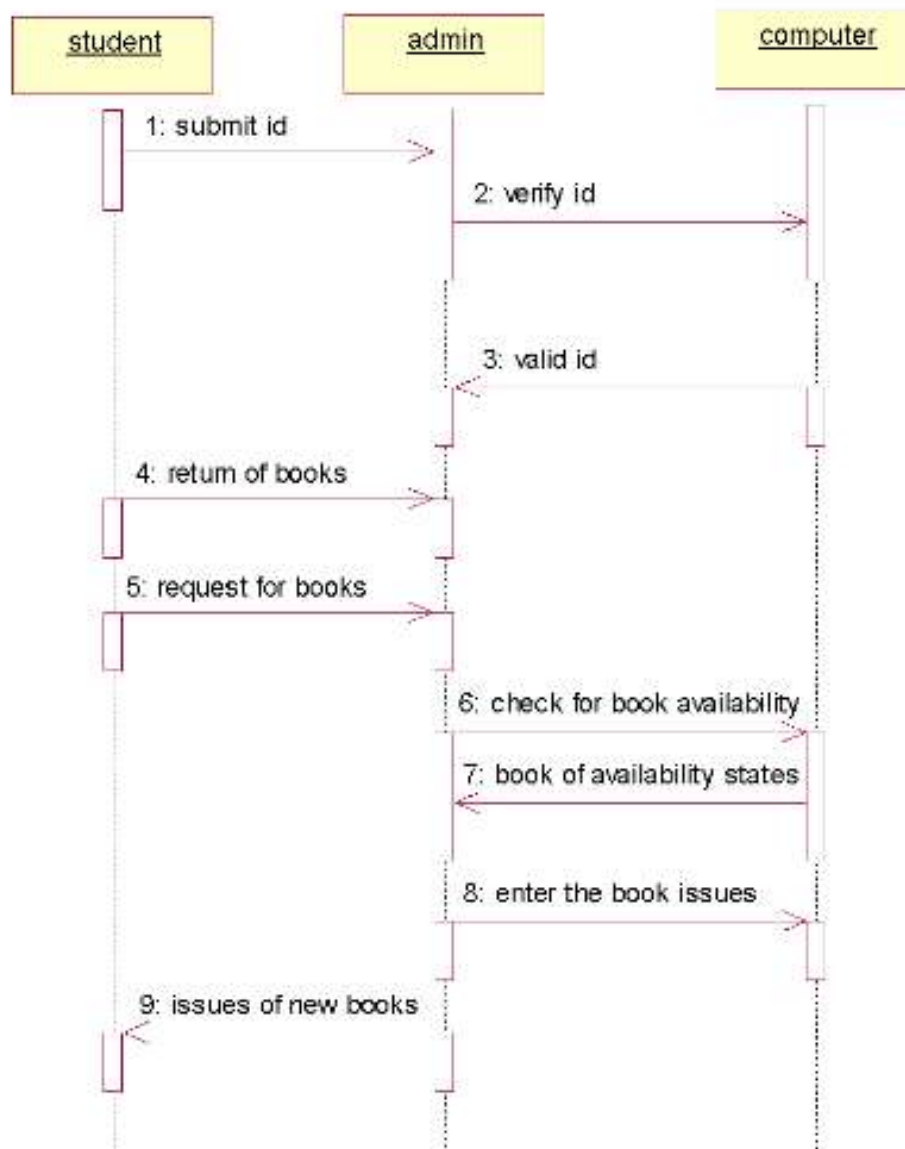
Decision box: Check availability of books whether it is present or not.



SEQUENCE DIAGRAM:

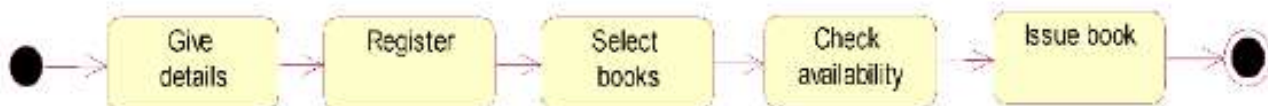
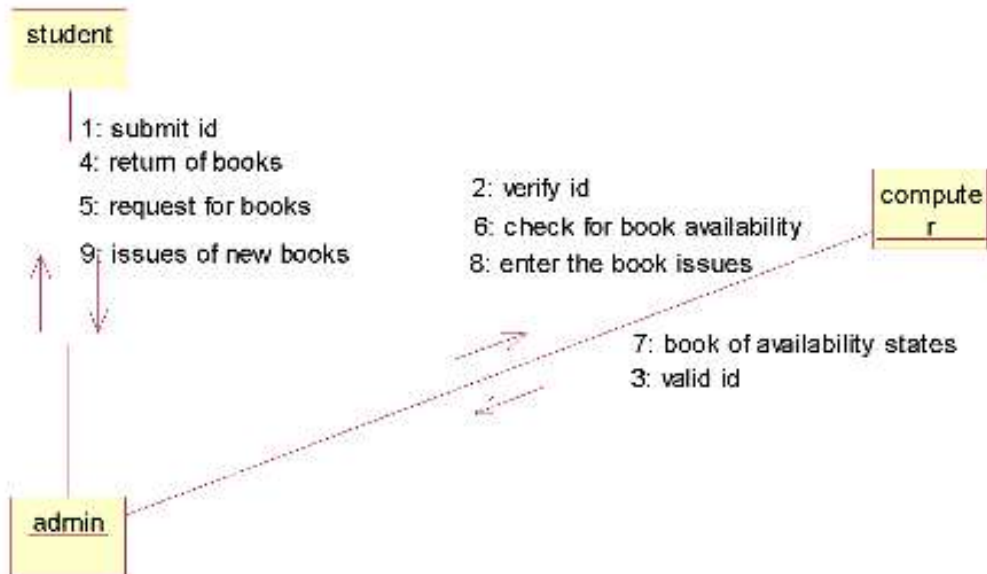
This diagram consists of the objects, messages and return messages.

Object: Stud, admin, computer



COLLABORATION DIAGRAM:

This diagram contains the objects and actors. This will be obtained by the completion of the sequence diagram and pressing the F5 key.

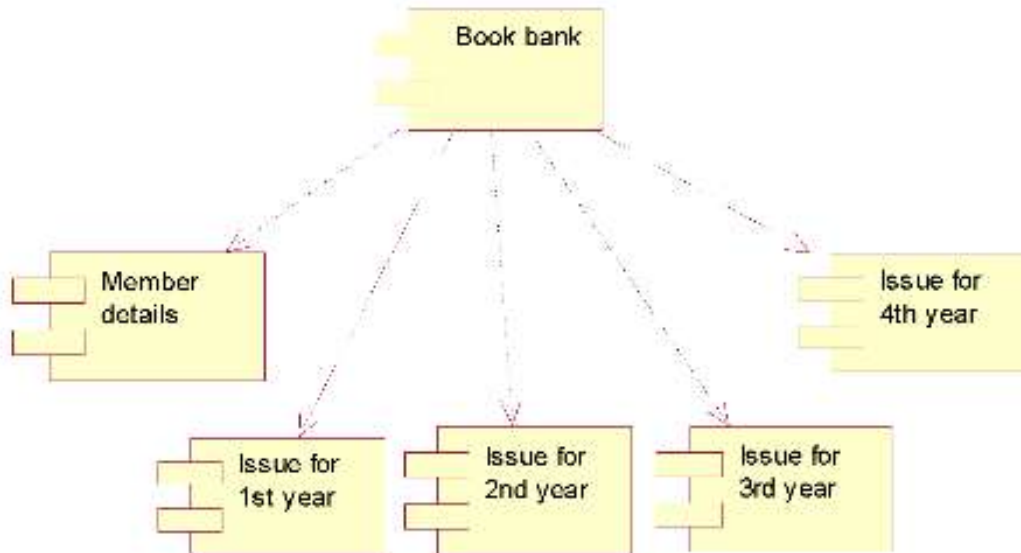


STATE CHART DIAGRAM

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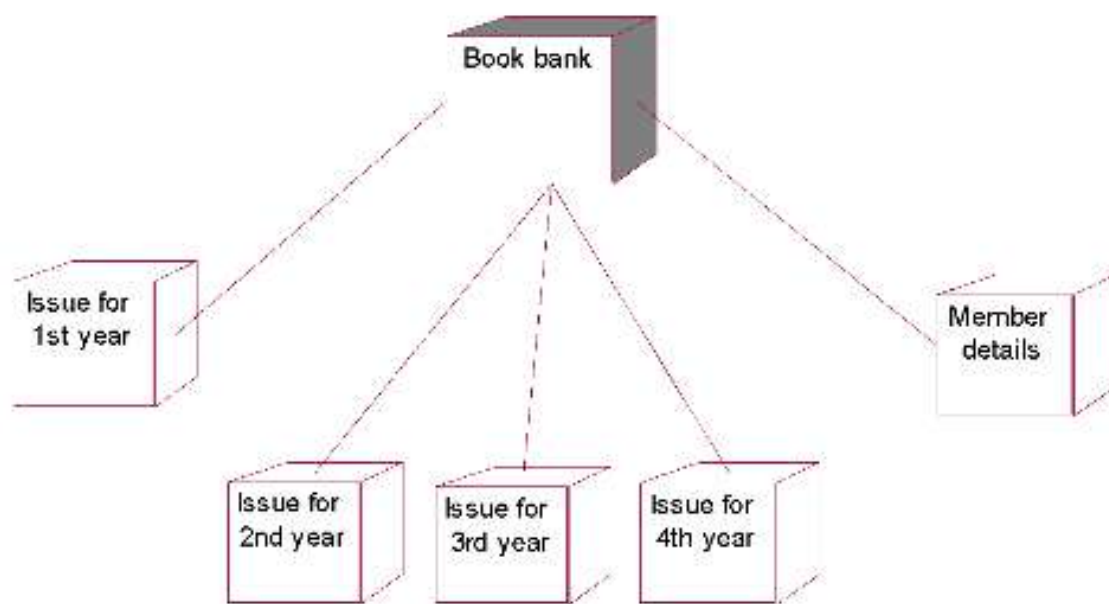
COMPONENT DIAGRAM

The component diagram's main purpose is to show the structural relationships between the components of a system. It is represented by boxed figure. Dependencies are represented by communication association



DEPLOYMENT DIAGRAM

A deployment diagram in the unified modeling language serves to model the physical deployment of artifacts on deployment targets. Deployment diagrams show "the allocation of artifacts to nodes according to the Deployments defined between them. It is represented by 3- dimensional box. Dependencies are represented by communication association



RESULT:

Thus the diagrams [use case, activity, sequence, collaboration, class, collaboration, component, deployment, package] for the Book bank registration system has been designed, executed and output is verified.